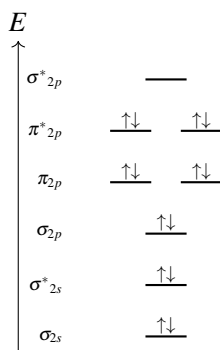


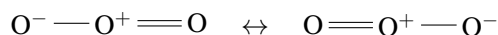
- the S–S bond in hydrogen disulfide (H_2S_2). $\therefore$ the Lewis structure, including any resonance contribution where applicable, must first be used to determine the *effective amount of bonding electron density shared between the nuclei*. Smaller bonding electron density weakens the attractive interaction and places the potential-energy minimum at a larger internuclear separation, producing the longest equilibrium bond.
- Emitted $\therefore$ a photon is emitted when an electron transitions to a lower energy level
- 8 valence electrons $\therefore$ sum the group valence electrons of every atom; 2 lone pairs on the central O atom
- Li^+ is a cation. It has 3 protons and 2 electrons.
- Zr, Pd, Ag, Sn $\therefore$ elements with greater metallicity hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
- $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
- 8.559 g
- 3.23 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
- Oxidizing Agent: H_3AsO_4 Reducing Agent: Be
- $M = \frac{\rho \cdot N_A \cdot a^3}{Z} = \frac{2.7 \times 6.022 \times 10^{23} \times (404.95 \times 10^{-10})^3}{4} = 27.0 \text{ g/mol}$ $\therefore$ the metal is aluminum (Al).
- The σ_{2p} orbital lies lower $\therefore$ O_2 has no s – p mixing, so the normal ordering (σ_{2p} below π_{2p}) applies.
- $\text{A} \rightarrow \text{C}$ $\therefore$ because this transition corresponds to the largest energy difference, resulting in absorption of a photon with the highest frequency (and shortest wavelength).
- Sample A $\therefore$ $\rho_A = 8.13 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 0.537 \frac{\text{g}}{\text{mL}}$
- $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
- $2.78 \times 10^{10} \text{ g}$
- $6.57 \times 10^{-7} \text{ m}$
- $n_{\text{met}} = \frac{6.40 \text{ g}}{32.04} = 0.200 \text{ mol}$; $n_{\text{H}_2\text{O}} = \frac{90.0 \text{ g}}{18.015} = 5.00 \text{ mol}$; $\chi_{\text{met}} = 0.0384$; $\chi_{\text{H}_2\text{O}} = 0.962$; $\chi_{\text{met}} + \chi_{\text{H}_2\text{O}} = 1.00$ ✓
- 11 cm^3 . $\therefore$ at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:1 ratio of PCl_5 to Cl_2 , so the product volume is multiplied by 1/1.
- dinitrogen pentoxide
- KCl: $i \times M = 2 \times 0.200 \text{ M} = 0.400 \text{ osmol/L}$ (hypertonic); Na_2SO_4 : $i \times M = 3 \times 0.100 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic); $\text{C}_{12}\text{H}_{22}\text{O}_{11}$: $i \times M = 1 \times 0.250 \text{ M} = 0.250 \text{ osmol/L}$ (hypotonic). $\therefore$ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.
- Bond order = 0.5; paramagnetic; stable $\therefore$ there are 1.0 more bonding than antibonding electrons and the MO configuration contains 1 unpaired electron.
- sp^2 for every carbon. Each C in the ring forms 2 C–C bonds and 1 C–H bond (3 electron groups) $\therefore$ all C atoms are sp^2 -hybridized. One unhybridized p orbital per C overlaps to form a delocalized π system above and below the ring.
- sp^3 $\therefore$ the number of regions of electron density equals the number of hybrid orbitals required around the atom.
- $7.2 \times 10^{-8} \text{ m}$
- $\rho = \frac{Z \cdot M}{N_A \cdot a^3} = \frac{4 \times 78.075}{6.022 \times 10^{23} \times (546.3 \times 10^{-10})^3} = 3.18 \text{ g/mL}$
- The bond dissociation energy (151 kJ/mol) $\therefore$ this is the energy required to separate the bonded atoms from the minimum of the potential-energy well to the noninteracting state at infinite separation.
- The following compounds have a dipole moment: NO
- 4.490 km
- (lowest energy) red light < orange light < yellow light < blue light (highest energy)
- 3.28 g

31. Hybridization: sp^2 . Electron-Group Geometry: trigonal planar. Molecular Geometry: trigonal planar.
 $\therefore$ there are 3 bonds and 0 lone pairs around the central atom, there are 3 total electron groups surrounding it and $\therefore$ each requires one hybrid orbital, the central atom is sp^2 hybridized. The electron-group geometry is trigonal planar. Because all electron groups are bonding pairs, the molecular geometry is also trigonal planar with bond angles of 120° .
32. σ/π bonding. Restricted rotation arises because rotating a π bond destroys the required side-by-side overlap of p orbitals, whereas a σ bond can rotate while maintaining end-to-end overlap.
33. 2.35×10^1 mol
34. 34.7 mL
35. LiCl $\therefore$ greatest freezing point depression $\therefore$ largest effective molality. $C_6H_{12}O_6$: $i \times m = 1 \times \frac{1.00}{180.16} = 5.55 \times 10^{-3}$ mol/kg;
 $AlCl_3$: $i \times m = 4 \times \frac{1.00}{133.34} = 0.0300$ mol/kg; LiCl: $i \times m = 2 \times \frac{1.00}{42.39} = 0.0472$ mol/kg.
36. $5O_2 + P_4 + 6H_2O \rightarrow 4H_3PO_4$
37. $A \rightarrow C$ $\therefore$ this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.
38. 16.2 cm^3 . $\therefore$ at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:4 ratio of C_2H_2 to CO_2 , so the product volume is multiplied by 4/2.
39. Bond order = $\frac{8-6}{2} = 1$; diamagnetic (0 unpaired electrons). There are 14 valence electrons. The molecular orbitals are ordered using the correct mixing model and populated according to the Aufbau principle, the Pauli exclusion principle, and Hund's rule. The bond order follows from $\frac{N_b - N_a}{2}$, and the magnetic behavior follows from the presence or absence of unpaired electrons in the completed molecular-orbital diagram.



40. molecular
41. The solution density. $\therefore$ molarity is per liter of solution but molality is per kilogram of solvent; density converts the solution volume into mass so the solvent mass can be found.
42. No. $\therefore$ combining atomic orbitals produces *new linear combinations* with *different spatial distributions*, so the hybrid orbitals have shapes distinct from the original s and p orbitals while spanning the same orbital space.
43. 3.29 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
44. Saturated $\therefore$ dissolution and crystallization are in dynamic equilibrium.
45. Increasing pressure lowers the melting point $\therefore$ pressure favors the denser liquid (smaller molar volume), so the solid-liquid boundary has a negative slope.
46. $S_8 + 12O_2 + 8H_2O \rightarrow 8H_2SO_4$
47. Strong electrolyte $\therefore$ strong base; dissociates completely: $K^+ + OH^-$.
48. The student assigning sp^2 is correct. Every valid Lewis structure for this species has the same number of electron domains around the central atom (3), so the σ framework and electron-domain geometry are unchanged. Because hybridization follows from that geometry rather than the placement of π bonds, the assignment is identical for every valid Lewis structure.
49. 6.079×10^1 MPa
50. 57.2%
51. 0.93 mL of ammonium dichromate

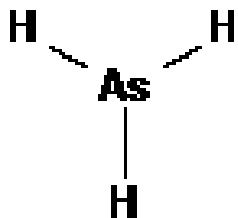
52. No. The atoms are connected differently in the two structures. Changing one representation into the other requires breaking existing covalent bonds and forming new ones because the molecular connectivity is different.
53. 31.5 mL
54. Sublimation only (solid $\rightarrow$ gas) $\therefore$ 0.003 atm is below H₂O's triple-point pressure (0.006 atm), so the solid converts directly to vapor.
55. The SI base unit for mass is the kilogram (symbol: kg).
56. 0.46. $\therefore$ 25 g of potassium chlorate corresponds to 0.20 mol. The reaction produces 0.31 mol of oxygen. The amount of neon remains 0.260 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
57. Yes; final temperature = -86.°C. $\therefore$ the amount of gas and volume remain constant, the ideal gas law requires pressure and absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.
58. Bond order = 1.5 $\therefore$ there are 3.0 more bonding than antibonding electrons.
59. Endothermic ($\Delta H_{\text{sol}} = +3.9 \text{ kJ/mol}$) $\therefore$ solution temperature will decrease. $\therefore$ slightly endothermic; entropy increase still drives dissolution at room temperature.
60. $3.53 \times 10^{-22} \text{ g}$
61. 2.0142 mol
62. 3p: $n = 3, l = 1, N_e = 6 \dots$ 4d: $n = 4, l = 2, N_e = 10 \dots$ 3d: $n = 3, l = 2, N_e = 10$
63. 7.245 g
64. Consistent. $\therefore$ Combining the measured density with the ideal gas law gives the experimental molar mass, which matches the molar mass of He.
65. nitrogen trifluoride
66. 2 σ bonds and 2 π bonds. The Lewis structure contains one H–C single bond and one C $\equiv$ N triple bond. Every bonded pair contributes one σ bond, while the triple bond contributes two additional π bonds.
67. Oxidized: Li Reduced: Sr
68. 60zJ $\therefore$ "A" is the level with the lowest energy, AKA the ground state.
69. sp^3 ; tetrahedral electron-group geometry; tetrahedral molecular geometry; 109.5° $\therefore$ the Lewis structure contains 4 electron groups around the central atom. 4 electron groups require a tetrahedral arrangement, so the central atom uses sp^3 hybrid orbitals to orient those electron groups for minimum repulsion. The arrangement of bonding pairs and lone pairs within that electron-group geometry produces the observed molecular geometry and corresponding bond angle(s).
70. (lowest frequency) microwaves < orange light < green light < gamma rays (highest frequency)
71. 0.0830 mol
72. $\Delta H_{\text{vap}} = 2260 \text{ J/g} \times 18.015 \text{ g/mol} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 40.7 \text{ kJ/mol}$
73. Fe
74. 7.44 g
75. 97.0 g
76. Solution A: $\Delta T = i K m = 2 \times 0.512 \times 0.200 = 0.205^\circ \text{C}$; Solution B: $\Delta T = 1 \times 0.512 \times 0.200 = 0.102^\circ \text{C}$. Solution A has the greater boiling point elevation $\therefore$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).
77. It will be depressed (fall) $\therefore$ adhesion to wax (paraffin) is weaker than cohesion, so the liquid is pushed down.
78. N
79. $\text{RbI} < \text{NaCl} < \text{MgO}$ $\therefore$ L.E. increases with greater ionic charge and smaller ionic radius (Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$)
- 80.



the bond order of each O–O bond is 1.5 $\therefore$ the bond order of a bond in a resonance hybrid is the average of its bond orders in the equivalent resonance structures; all of the resonance structures contribute equally, so every O–O bond is identical in the resonance hybrid. Sum the bond order of each O–O bond in O₃ and divide by the number of equivalent resonance structures.

81. 0.330. $\therefore$ the cylinder volume does not change, reducing the pressure from the initial value to atmospheric pressure reduces the amount of gas in direct proportion. The remaining fraction equals the final pressure divided by the initial pressure.

82. H_2O (0.031 atm) < $\text{C}_2\text{H}_5\text{OH}$ (0.056 atm) < $(\text{C}_2\text{H}_5)_2\text{O}$ (0.726 atm) $\therefore$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O : strong H-bonding; strongest IMF; lowest VP; $\text{C}_2\text{H}_5\text{OH}$: H-bonding; medium IMF; $(\text{C}_2\text{H}_5)_2\text{O}$: LDF only; low MM; weakest IMF).
83. 1 $\therefore$ 1 O–H donor + 2 lone pairs on O acceptors.
84. 90. kPa. $\therefore$ the gas sample and its temperature do not change, PV is constant. The smaller final volume produces a proportionally larger pressure, so the initial pressure must be lower by the inverse volume ratio.
85. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. $\therefore$ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.



86. NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.
87. KNO_3 crystallizes from solution on cooling. $\therefore$ KNO_3 solubility increases with T; cooling creates supersaturation $\rightarrow$ crystallization.
88. Copper most commonly forms the cation Cu^+ .
89. 0.226 M
90. -14 J
91. $9.533 \times 10^0\text{ g}$
92. e^- -group arrangement: Linear
Shape: Linear
Ideal bond angle(s): 180°
93. $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
94. CCl_4 (77°C) > SiCl_4 (58°C) > CS_2 (46°C) > CO_2 (-78°C). $\therefore$ All are essentially nonpolar, so London dispersion forces dominate. Larger and more polarizable electron clouds produce stronger dispersion forces and therefore higher boiling points.
95. Mole fraction. $\therefore$ it is moles of one component over total moles, it is dimensionless and temperature-independent, which makes it ideal for vapor-pressure (Raoult's law) calculations.
96. -4°F
97. 167% of the initial volume. $\therefore$ rearranging the combined gas law gives $V_2/V_1 = (T_2/T_1)(P_1/P_2)$. The final volume is determined by the competing effects of the change in absolute temperature and the change in pressure.
98. The additional bond is lost $\therefore$ a π bond requires parallel side-by-side overlap of p orbitals, and rotating the orbitals removes that overlap so the stabilizing electron density between the nuclei disappears.
99. ionic, base
100. Basis: 100 g solution $\rightarrow$ 20.0 g HCl, 80.0 g water; $n_{\text{solute}} = \frac{20.0\text{ g}}{36.46\text{ g/mol}} = 0.549\text{ mol}$; $m = \frac{0.549\text{ mol}}{0.0800\text{ kg}} = 6.86\text{ mol/kg}$
101. The π_{2p} orbitals lie lower $\therefore$ B_2 shows s - p mixing, which raises σ_{2p} above π_{2p} .
102. -1
103. Supersaturated $\therefore$ unstable; crystallization occurs on seeding or disturbance.
104. Hg (485.5 mN/m) > H_2O (72.8 mN/m) > C_6H_{14} (18.4 mN/m) > CH_4 (13.0 mN/m) $\therefore$ surface tension $\propto$ IMF strength; Hg : metallic bonding; extremely strong cohesion; H_2O : extensive hydrogen bonding; C_6H_{14} : London dispersion forces only; CH_4 : very weak dispersion forces.
105. Covalent-network solid (diamond) $\therefore$ covalent-network; no free charges; covalent bonds throughout lattice.
106. $3.33 \times 10^4\text{ Pa}$. $\therefore$ the balanced equation determines the amount of O_2 produced from the amount of H_2O_2 that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
107. 0.43 atm. $\therefore$ pressure in an ideal gas is proportional to the amount present at fixed temperature and volume, removing only chlorine removes only its contribution to the total pressure while the other gas is unchanged.

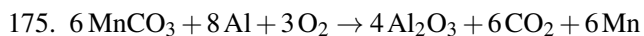
108. $i = \frac{\Delta T}{K_f m} = \frac{0.372}{1.86 \times 0.100} = 2.00$; the solute is dissociating completely into 2 ions, matching the ideal $i = 2$.
109. 1 $\therefore$ 3 N–H bonds (donor) + 1 lone pair on N (acceptor).
110. $\text{Fe}^{3+}(\text{aq}) + 3\text{F}^{-}(\text{aq}) \rightarrow \text{FeF}_3(\text{s})$
111. 66.3 kJ
112. 90 zJ $\therefore$ "A" is the level with the lowest energy, AKA the ground state.
113. $1.09 \times 10^2 \text{ g}$
114. 7.68 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the $\text{H}_2\text{O}:\text{O}_2$ coefficient ratio of 2.5:1 determines the required volume.
115. $7.61 \times 10^1 \text{ Gm}$
116. Endothermic ($\Delta H_{\text{sol}} = +10.9 \text{ kJ/mol}$) $\therefore$ solution temperature will decrease. $\therefore$ hydrogen bonding with water is slightly less favorable than solute-solute H-bonds.
117. Melting then vaporization (solid $\rightarrow$ liquid $\rightarrow$ gas) $\therefore$ 1.0 atm exceeds the triple-point pressure, so the solid melts (114°C) then boils (184°C).
118. 0.035 J/g/K
119. 44 g/mol . $\therefore$ density gives the mass per unit volume, while the ideal gas law relates volume to the number of moles, combining these relationships yields the mass of one mole of the gas.
120. No new substance is formed $\therefore$ The wire changes shape but remains copper.
121. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +853 + (-890) = -37.0 \text{ kJ/mol}$; exothermic $\therefore \Delta H_{\text{sol}} < 0$.
122. 15 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
123. $8.642 \times 10^0 \text{ g}$
124. $9.13 \times 10^{-7} \text{ m}$
125. Decreases. $\therefore$ the pressure remains constant, heating causes the gas volume to increase while the amount of gas is unchanged. The same mass therefore occupies a larger volume, decreasing the density.
126. 14.1 mL . $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{O}_2:\text{CO}$ is 1:2 so the volume changes by the same factor.
127. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
128. 15.0 g
129. 4237.1 g
130. Yes, Tyndall effect is observed. Colloid $\therefore$ proteins and cells produce a Tyndall effect; plasma proteins are colloidal.
131. 268.65 K
132. 23.05 g
133. $\text{C} \rightarrow \text{B}$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
134. $2\text{NH}_4\text{F} + \text{Ba}(\text{MnO}_4)_2 \rightarrow \text{BaF}_2 + 2\text{NH}_4\text{MnO}_4$
135. sp^3d hybridization. $\therefore$ The chlorine has three bonding groups and two lone pairs, giving five electron groups. Five electron groups adopt a trigonal bipyramidal electron-domain arrangement requiring sp^3d hybridization; the two equatorial lone pairs leave a T-shaped molecular geometry with bond angles slightly less than 90° . $\therefore$ the molecular geometry is T-shaped with an ideal bond angle of $<90^\circ$.
136. He_2 : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*2}$; bond order $= \frac{2-2}{2} = 0$; diamagnetic (not stable).
 He_2^+ : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*1}$; bond order $= \frac{2-1}{2} = 0.5$; paramagnetic.
 Removing one electron changes the electron occupancy of the highest-energy MO. Adding an electron to an antibonding orbital *destabilizes* the molecule, *lowering* the bond order, while removing one from an antibonding orbital *stabilizes* the molecule, raising the bond order. An unpaired electron makes the species paramagnetic; if all electrons are paired, it is diamagnetic.
137. $5\text{O}_2 + \text{P}_4 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
138. 75%

139. 0.099 J/g/K
140. $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 62.36 - 19.06 = 43.3 \text{ kJ/mol}$
141. -8 J
142. $7.02 \times 10^0 \text{ mol}$
143. $0.022 \frac{\text{mol}}{\text{L}}$
144. 3454 kJ $\therefore$ forming bonds is exothermic; sum the bond energies of every product bond
145. Molecular solid (I_2) $\therefore$ neutral molecules held by weak IMF; low lattice energy $\rightarrow$ low melting point.
146. $\text{Sb}(s) + 3 \text{Au}^+(aq) \rightarrow \text{Sb}^{3+}(aq) + 3 \text{Au}(s)$
147. 17 protons and 18 neutrons.
148. Physical property. The density of lead is a physical property; it can be measured without changing the chemical identity of lead.
149. 1.3 mL
150. $\chi_{\text{H}_2\text{O}} = \frac{9.99}{9.99 + 0.0999} = 0.990$; $P = 0.990 \times 355.1 \text{ mmHg} = 352 \text{ mmHg}$; $\Delta P = 3.52 \text{ mmHg}$ lowering.
151. 2 σ bonds and 2 π bonds. The Lewis structure contains two C=O double bonds. Each double bond contributes one σ bond from end-to-end overlap and one π bond from side-by-side overlap, giving two of each overall.
152. 974000000000 Hz
153. $7.3 \times 10^{-9} \text{ g}$
154. Ammonia has the higher boiling point. $\therefore$ its hydrogen atoms are bonded directly to nitrogen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.
155. Both students must assign the same hybridization: sp^2 . The hybridization is determined by the number of electron domains around the central atom (3), which fixes the σ framework. Moving a double bond changes only the distribution of π electrons and formal charges, so the electron-domain geometry and hybridization remain unchanged.
156. $4.4950 \times 10^{-3} \text{ moles}$
157. Graphite functions better as a dry lubricant. $\therefore$ graphite contains many stacked carbon sheets held together by weak intermolecular attractions that allow adjacent layers to slide over one another. Graphene is a single sheet and therefore lacks stacked layers that can slide.
158. 40.N Only the piston area is needed because pressure is force per unit area; the gas height does not affect the force once the pressure is known. $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
159. $m = \frac{0.402 \text{ mol}}{0.200 \text{ kg}} = 2.01 \text{ mol/kg}$; $\Delta T_f = i K_f m = 2 \times 1.86 \times 2.01 = 7.48^\circ\text{C}$; $T_f = 0.00 - 7.48 = -7.48^\circ\text{C}$
160. Decomposition
161. $\text{CH}_4\text{N}_2\text{O}$ ($i = 1$) < MgBr_2 ($i = 3$) < Na_2SO_4 ($i = 3$) $\therefore$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. $\text{CH}_4\text{N}_2\text{O}$: nonelectrolyte; 1 particle; MgBr_2 : strong electrolyte; $\text{Mg}^{2+} + 2\text{Br}^- \rightarrow 3$ particles; Na_2SO_4 : strong electrolyte; $2\text{Na}^+ + \text{SO}_4^{2-} \rightarrow 3$ particles.
162. 0.731 m
163. Water flows from the 0.10 M solution into the 0.40 M solution. $\therefore$ water moves toward the side with higher osmotic pressure.
164. $3.0080 \times 10^3 \text{ m}$
165. $m = \frac{0.139 \text{ mol}}{0.250 \text{ kg}} = 0.555 \text{ mol/kg}$; $\Delta T_b = i K_b m = 1 \times 0.512 \times 0.555 = 0.284^\circ\text{C}$; $T_b = 100.0 + 0.284 = 100.^\circ\text{C}$
166. 3.36 m
167. $2\text{H}_2\text{O}(l) + 2\text{Cs}(s) \rightarrow \text{H}_2(g) + 2\text{CsOH}(aq)$
168. 0.48% $\therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m \Delta v$
169. $\text{A} \rightarrow \text{B}$ $\therefore$ because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).
170. $1.408 \times 10^{18} \text{ moles}$
171. O_2^- : $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \pi_{2p_x}^2 \pi_{2p_y}^2 \pi_{2p_x}^{*2} \pi_{2p_y}^{*1}$; unpaired MO(s): a π_{2p}^* orbital; paramagnetic.
172. 98.0 g/mol

173. $\text{NaCl} (-787 \text{ kJ/mol}) < \text{CaO} (-3414 \text{ kJ/mol}) < \text{MgO} (-3795 \text{ kJ/mol})$ $\therefore$ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \frac{q_+q_-}{r}$).



174.



176. $\text{B} \rightarrow \text{C}$ $\therefore$ because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).

177. Ionic solid $\therefore$ its composition implies a very high melting point ($2+/2-$ ions), brittleness, and conduction only when molten.

178. $\Delta H_{\text{vap}} = \frac{30.8 \text{ kJ/mol}}{78.114 \text{ g/mol}} = 0.394 \text{ kJ/g}$

179. $M = \frac{\Pi}{RT} = \frac{4.30 \times 10^{-3} \text{ atm}}{0.08206 \times 298 \text{ K}} = 1.76 \times 10^{-4} \text{ mol/L}; \mathcal{M} = \frac{m/V}{M} = \frac{2.50 \text{ g/L}}{1.76 \times 10^{-4} \text{ mol/L}} \approx 1.42 \times 10^4 \text{ g/mol}$

180. $2.1447 \times 10^2 \text{ nm}$

181. metallic; high melting point $\therefore$ its cations are held together by a sea of delocalized electrons

182. Solubility decreases $\therefore$ gas solubility always decreases with increasing temperature.

183. The emulsifier is amphiphilic: its nonpolar tail dissolves in each oil droplet while its polar head faces the water, coating the droplets so they cannot coalesce and separate.

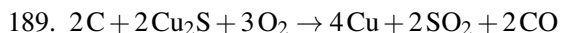
184. It is a compound. $\therefore$ it is a pure substance formed when two or more different chemical elements bond together in a fixed ratio.

185. Graphite functions better as a dry lubricant. $\therefore$ graphite contains many stacked carbon sheets held together by weak intermolecular attractions that allow adjacent layers to slide over one another. Graphene is a single sheet and therefore lacks stacked layers that can slide.

186. 0.90 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.

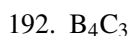
187. $\text{D} \rightarrow \text{C}$ $\therefore$ because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).

188. It is the equilibrium separation between molecules $\therefore$ at the minimum the net force is zero (attraction balances repulsion), and molecules in a solid vibrate about this distance.



190. 5 σ bonds and 1 π bond. The Lewis structure contains four C–H single bonds and one C=C double bond. Every bonded pair contributes one σ bond, while the double bond contributes one additional π bond.

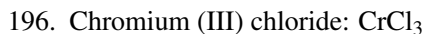
191. $7.0 \times 10^{-8} \text{ m}$



193. Answer: X belongs to Group 17 $\therefore$ X is Br-like and has 7 valence electrons.

194. 27 mM

195. Single-Displacement



197. $M = \frac{\rho \cdot N_A \cdot a^3}{Z} = \frac{19.25 \times 6.022 \times 10^{23} \times (316.52 \times 10^{-10})^3}{2} = 183.8 \text{ g/mol} \therefore$ the metal is tungsten (W).

198. 32

199. $8.6 \frac{\text{m}}{\text{s}}$

200. London dispersion and dipole–dipole $\therefore$ weakly polar; P–H does not hydrogen-bond.

201. $\text{C}_2 < \text{C}_2^- < \text{C}_2^+$. C_2 : $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p_x}^2 \pi_{2p_y}^2$; unpaired MO(s): none; diamagnetic. C_2^- : $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p_x}^2 \pi_{2p_y}^2 \sigma_{2p}^1$; unpaired MO(s): σ_{2p} ; paramagnetic. C_2^+ : $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p_x}^2 \pi_{2p_y}^1$; unpaired MO(s): a π_{2p} orbital; paramagnetic. Species with more unpaired electrons are more strongly attracted to a magnetic field.

202. -787 kJ $\therefore$ Hess's law: $\Delta H_f^\circ = \Delta H_{\text{sub}} + IE + \Delta H_{\text{diss}} + EA + \Delta H_{\text{lat}}^\circ$

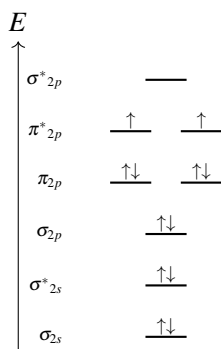
203. O: sp^3 — justified by 2 bonds and 2 lone pairs

204. It will be depressed (fall) $\therefore$ adhesion to copper is weaker than cohesion, so the liquid is pushed down.
205. $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 43.20 - 4.6 = 38.6 \text{ kJ/mol}$
206. NO
207. $+0 \therefore FC = \text{valence } e^- - \text{unshared } e^- - \frac{1}{2} \text{shared } e^- \therefore FC = 4 - 0 - \frac{1}{2}(8)$
208. 0.28 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
209. 11 σ bonds and 1 π bond. The π bond comes from the C=C bond. $\therefore$ Counting the σ bond in every covalent bond and the additional π bond(s) present only in the multiple bond(s) gives these totals.
210. -2
211. $5.08 \times 10^{-4} \text{ g}$
212. 22.9%. $\therefore$ the fertilizer's nitrogen content is the sum of the nitrogen supplied by each ingredient, so each ingredient's contribution equals (mass percent of ingredient) $\times$ (mass fraction of nitrogen in that compound), and these contributions are added together.
213. 35 N The piston is square, so its area is side². $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
214. 146 mL
215. $m = \frac{\Delta T_b}{i K_b} = \frac{1.02}{1 \times 0.512} = 2.00 \text{ mol/kg}$; $n = m \times m_{\text{solvent}} = 2.00 \times 0.200 \text{ kg} = 0.400 \text{ mol}$; $m_{\text{solute}} = 0.400 \text{ mol} \times 180.16 \text{ g/mol} = 72.1 \text{ g}$
216. 1.3 atm. $\therefore$ the amount of the second gas is the difference between the total amount and the known amount, and each gas produces pressure according to its own amount through $P = nRT/V$.
217. $2.23 \times 10^1 \text{ g}$
218. The experimentally observed bond angle(s) are inconsistent with the orientations of unmodified atomic orbitals $\therefore$ hybridization mixes the O atom's s and p orbitals to produce sp^2 hybrid orbitals whose orientations are consistent with the observed geometry.
219. 5.00 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:1 particle ratio between HBr and H_2 , so their volumes must be in the same ratio.
220. $q = mc\Delta T = 26.0 \times 4.184 \times 55 = 5.98 \text{ kJ}$
221. Pure substances: A, B $\therefore$ the mass ratio of E_1 to E_2 in A and B is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
222. $A \rightarrow G \therefore$ because this transition corresponds to the largest energy difference, resulting in absorption of a photon with the highest frequency (and shortest wavelength).
223. AgI and H_2O
224. A_2B_4
225. It will sublime $\therefore$ the external pressure (2.4568 atm) is below the triple-point pressure (5.11 atm), so no liquid is stable and the solid converts directly to gas.
226. 2.1% $\therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
227. Cs, Rb, K, Na $\therefore$ elements with greater metallicity hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
228. 59.3%
229. All have a bond order of 1. $\therefore$ they are isoelectronic, they occupy the same molecular orbitals with the same electron configuration, producing the same numbers of bonding and antibonding electrons and therefore the same bond order.
230. Answer: X belongs to Group 1 $\therefore$ X is Cs-like and has 1 valence electrons.
231. The following compounds have a dipole moment: NF_3 , H_2S , HCN
232. Pure substances: A, C $\therefore$ the mass ratio of E_1 to E_2 in A and C is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
233. $m = \frac{0.111 \text{ mol}}{0.200 \text{ kg}} = 0.555 \text{ mol/kg}$; $\Delta T_f = i K_f m = 1 \times 1.86 \times 0.555 = 1.03^\circ\text{C}$; $T_f = 0.00 - 1.03 = -1.03^\circ\text{C}$

234. 0.760 moles
235. Element $\therefore$ it contains only phosphorus atoms.
236. metallic $\therefore$ it is a pure metal whose atoms are held together by metallic bonding involving delocalized valence electrons.
237. 19.2 mL
238. $0.4 \frac{\text{mol}}{\text{L}}$
239. $m = \frac{\Delta T_b}{iK_b} = \frac{0.256}{1 \times 0.512} = 0.500 \text{ mol/kg}$; $n = m \times m_{\text{solvent}} = 0.500 \times 0.100 \text{ kg} = 0.0500 \text{ mol}$; $m_{\text{solute}} = 0.0500 \text{ mol} \times 342.3 \text{ g/mol} = 17.1 \text{ g}$
240. $7.81 \times 10^1 \text{ Gm}$
241. $5.9 \times 10^{-24} \text{ moles}$
242. 6 $\therefore$ each atom touches 4 neighbors in its plane, 1 above, 1 below.
243. $M = \frac{\rho \cdot N_A \cdot a^3}{Z} = \frac{2.7 \times 6.022 \times 10^{23} \times (404.95 \times 10^{-10})^3}{4} = 27.0 \text{ g/mol}$ $\therefore$ the metal is aluminum (Al).
244. 6.837 g
245. $5.249 \times 10^0 \text{ km}$
246. $3.8 \times 10^{-8} \text{ m}$
247. 0.386 M
248. Element
249. $\text{H}_2\text{S} + 2\text{RbOH} \rightarrow 2\text{H}_2\text{O} + \text{Rb}_2\text{S}$
250. O_2^+ is paramagnetic and weakly attracted $\therefore$ its 1 unpaired electron give a net magnetic moment that aligns with the field.
251. 0.11 m. $\therefore$ Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through $A = F/P$. Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
252. Random error $\therefore$ the measurements fluctuate unpredictably in both directions.
253. Water flows from the 0.05 M solution into the 0.20 M solution. $\therefore$ water moves from lower solute concentration to higher solute concentration.
254. $8.8 \times 10^{-11} \text{ g}$
255. 26 g/mol. $\therefore$ density gives the mass per unit volume, while the ideal gas law relates volume to the number of moles, combining these relationships yields the mass of one mole of the gas.
256. 4.474 kg
257. Dissolved N_2 comes out of solution (decompression sickness). $\therefore$ reduced pressure at surface lowers N_2 solubility (Henry's law); dissolved N_2 forms bubbles.
258. 98.0 g/mol
259. 2 $\therefore$ sp hybridization leaves 2 unhybridized p orbitals, and each can participate in one π bond.
260. 73.2 fm
261. $-1.37 \times 10^5 \text{ J}$
262. 17.61 g
263. London dispersion forces $\therefore$ both solute and solvent are nonpolar, so dispersion forces dominate.
264. $4\text{Al}(\text{OH})_3 + 3\text{C} \rightarrow 6\text{H}_2\text{O} + 4\text{Al} + 3\text{CO}_2$
265. $\text{S}_8 + 12\text{O}_2 + 8\text{H}_2\text{O} \rightarrow 8\text{H}_2\text{SO}_4$
266. Element $\therefore$ it contains only zinc atoms.
267. Intensive property. Melting point is an intensive property $\therefore$ it is the same regardless of how much of the substance is present.
268. Octahedral holes, coordination number 6 $\therefore$ $r_+/r_- = 102/181 = 0.56$ lies in the 0.414–0.732 range, which geometrically favors octahedral coordination.
269. 0.048 M
270. 0.234 m
271. 0.000087 m

272. The electron was added to an antibonding molecular orbital $\therefore$ occupying an antibonding molecular orbital decreases electron density between the nuclei by reinforcing destructive interference, weakening the attractive interaction and raising the total energy.
273. 0.36 m
274. No $\therefore$ the π electrons are delocalized equally over all six carbon atoms rather than confined to three individual double bonds. Equal electron sharing gives every carbon-carbon bond the same average bond order, so all six bonds have the same length.
275. Magnesium acetate: $\text{Mg}(\text{C}_2\text{H}_3\text{O}_2)_2$
276. σ/π bonding. The defining feature is how the orbitals overlap to form the bond. Hybridization determines which orbitals are available for bonding but does not classify the overlap as σ or π .
277. $1.10 \times 10^{-6} \text{ m}$ $\therefore$ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$
278. $q = n\Delta H_{\text{fus}} = \frac{19.9 \text{ g}}{58.443 \text{ g/mol}} \times 28.16 \text{ kJ/mol} = 9.59 \text{ kJ}$
279. -355 kJ . Since $\Delta H < 0$, the reaction is exothermic.
280. $6.66 \times 10^8 \text{ W}$
281. $\text{C}_{10}\text{H}_{20}$
282. London dispersion and dipole-dipole $\therefore$ the polar C-F bond gives a dipole, but the H atoms are on carbon (C-H does not hydrogen-bond).
283. H_2 . $\therefore$ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 2.02 g/mol. Only H_2 has that molar mass, so it is the only identity consistent with the measurements.
284. $3.2875 \times 10^{-3} \text{ moles}$
285. H_2O (72.8 mN/m) > $\text{C}_3\text{H}_8\text{O}_3$ (63.4 mN/m) > C_6H_6 (28.9 mN/m) > C_6H_{14} (18.4 mN/m) $\therefore$ surface tension $\propto$ IMF strength; H_2O : strong hydrogen bonding; $\text{C}_3\text{H}_8\text{O}_3$: extensive hydrogen bonding; C_6H_6 : large electron cloud; strong dispersion forces; C_6H_{14} : weaker dispersion forces.
286. It will melt $\therefore$ the external pressure (0.06 atm) is above the triple-point pressure (0.017 atm), so a stable liquid phase exists between solid and gas.
287. 0.0072 m
288. $q = n\Delta H_{\text{vap}} = \frac{38.1 \text{ g}}{18.015 \text{ g/mol}} \times 40.7 \text{ kJ/mol} = 86.1 \text{ kJ}$
289. Silver most commonly forms the cation Ag^+ .
290. 84.0%
291. Foam; e.g., shaving cream. $\therefore$ a gas-in-liquid dispersion defines this colloid type.
292. Not consistent. $\therefore$ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which does not match the molar mass of N_2O .
293. $3.79 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
294. 30%
295. 16 moles
296. 40 zJ $\therefore$ "A" is the level with the lowest energy, AKA the ground state.
297. $\text{A} \rightarrow \text{G}$ $\therefore$ this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.
298. $M = \frac{\Pi}{RT} = \frac{0.0111 \text{ atm}}{0.08206 \times 298 \text{ K}} = 4.54 \times 10^{-4} \text{ mol/L}$; $\mathcal{M} = \frac{m/V}{M} = \frac{30.0 \text{ g/L}}{4.54 \times 10^{-4} \text{ mol/L}} \approx 6.61 \times 10^4 \text{ g/mol}$
299. 21.2 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
300. 20.7 mol
301. 10 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
302. $5.35 \times 10^{11} \text{ Hz}$
303. $1.031 \times 10^5 \text{ Pa}$ $\therefore$ a barometer measures atmospheric pressure directly, so the reading is simply converted from torr to pascals.
304. $2.1 \frac{\text{mol}}{\text{L}}$

305. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^2 \pi_{2p}^2 \sigma_{2p}^2$
 Bond order = $\frac{8-2}{2} = 3$
 N_2 is diamagnetic. The bond order is 3 because there are 8 bonding and 2 antibonding electrons. It is diamagnetic because all electrons are paired.
306. 100 zJ $\therefore$ the energy of the absorbed photon is equal to the difference in energy levels. $|120 - 220| = 100$
307. 40. N The piston is rectangular, so its area is length $\times$ width. $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
308. Gas
309. polar covalent ($\Delta EN = 0.89$); O carries the δ^- $\therefore 0.4 \leq \Delta EN < 2.0$ and O is the more electronegative atom
310. molecular, weak acid
311. Supercritical fluid $\therefore$ both T and P exceed the critical point (-147°C , 34 atm).
312. ABAB, repeating $\therefore$ in hexagonal closest packing (HCP) the third layer sits directly above the first layer, distinguishing it from the other close-packed arrangement.
313. $1.35 \times 10^4 \text{ Pa}$ $\therefore$ a closed-end manometer has a vacuum at one end, so the height difference equals the absolute gas pressure before converting to SI units.
314. No new substance is formed $\therefore$ The wire changes shape but remains copper.
315. Diamagnetic; weakly repelled $\therefore$ the paired electron magnetic moments cancel, so the field can only induce a small opposing moment.
316. $4.2 \frac{\text{mol}}{\text{L}}$
317. $1.01 \times 10^5 \text{ Pa}$ $\therefore$ the total pressure equals the sum of the partial pressures of all gases present, including the water vapor, before converting to SI units.
318. $m = \frac{\Delta T}{K_b} = \frac{0.186}{0.512} = 0.363 \text{ mol/kg}$; $n = 0.363 \times 0.100 \text{ kg} = 0.0363 \text{ mol}$; $m_{anth} = 0.0363 \text{ mol} \times 178.23 \text{ g/mol} = 6.47 \text{ g}$
319. No. The atoms are connected differently in the two structures. Changing one representation into the other requires breaking existing covalent bonds and forming new ones because the molecular connectivity is different.
320. Melting all 37.0 g uses 12.4 kJ; the remaining heat warms the water $\therefore$ liquid water at $T = \frac{q - mH_{fus}}{mc} = 55.2^\circ\text{C}$.
321. 65°C
322. 0.31. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
323. Two layers form and vegetable oil is on top $\therefore$ vegetable oil is nonpolar and cannot hydrogen-bond with water, so they are immiscible; the less dense liquid floats, and density fixes only the stacking order, not whether they mix.
324. A_3B_5
325. the $\text{C}\equiv\text{N}$ bond in CN^- . $\therefore$ the Lewis structures, including any resonance contribution where applicable, must first be used to determine the effective amount of bonding electron density shared between the nuclei. Greater bonding electron density creates a deeper potential-energy well, so more energy is required to separate the atoms completely.
326. $\rho = \frac{Z \cdot M}{N_A \cdot a^3} = \frac{2 \times 183.84}{6.022 \times 10^{23} \times (316.5 \times 10^{-10})^3} = 19.3 \text{ g/mL}$
327. 343 K
328. $P_2 = P_1 \exp \left[-\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right] = 0.331 \text{ atm}$ $\therefore T_2 = 344.15 \text{ K}$, $T_1 = 373.15 \text{ K}$, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.
329. $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
330. $+1$ $\therefore FC = \text{valence } e^- - \text{unshared } e^- - \frac{1}{2} \text{shared } e^-$ $\therefore FC = 6 - 2 - \frac{1}{2}(6)$
331. Yes. The drawings differ only in the viewing direction of the same molecular arrangement. Changing the observer's perspective does not alter the connectivity or the bonding interactions between the atoms.
332. Bond order = $\frac{8-4}{2} = 2$; paramagnetic (2 unpaired electrons). There are 12 valence electrons. The molecular orbitals are ordered using the correct mixing model and populated according to the Aufbau principle, the Pauli exclusion principle, and Hund's rule. The bond order follows from $\frac{N_b - N_a}{2}$, and the magnetic behavior follows from the presence of unpaired electrons in the completed molecular-orbital diagram.



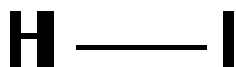
333. 10^{-3}

334. $1 \text{ } \cdot \cdot \text{ } 1$ O–H bond (donor) + 2 lone pairs on O (acceptor).

335. F $\cdot \cdot$ effective nuclear charge increases across a period, increasing an atom's attraction for bonding electrons

336. Methanol (CH_3OH). $\cdot \cdot$ Methanol molecules form hydrogen bonds through their O–H groups, whereas butane experiences only London dispersion forces. Stronger attractions make methanol more difficult to separate. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

337. 0.10 mol. $\cdot \cdot$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.



338. NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.

339. Strong electrolyte $\cdot \cdot$ ionic compound; dissociates completely: $\text{Na}^+ + \text{Cl}^-$.

340. 0.68. $\cdot \cdot$ 13 g of sodium sulfide corresponds to 0.17 mol. The reaction produces 0.17 mol of hydrogen sulfide. The amount of neon remains 0.352 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.

341. 8 $\cdot \cdot$ center atom touches all 8 corner atoms.

342. Oxidizing Agent: CO_2 Reducing Agent: K

343. $1.06 \times 10^5 \text{ Pa}$ $\cdot \cdot$ the mercury is higher on the open end, gas pressure exceeds atmospheric pressure, and the pressure difference is added to the atmospheric pressure before converting to SI units.

344. MO configuration: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^2 \pi_{2p}^2 \sigma_{2p}^1$

Bond order = $\frac{9-4}{2} = 2.5$ $\cdot \cdot$ bond order is determined by the balance between electrons in bonding (9) and antibonding (4) molecular orbitals. Bonding orbitals stabilize the molecule, whereas antibonding orbitals destabilize it, so a larger excess of bonding electrons produces a stronger bond.

N_2^+ is paramagnetic $\cdot \cdot$ its magnetic character is determined by whether any molecular orbitals contain unpaired electrons. One or more unpaired electrons produces paramagnetism, whereas complete electron pairing produces diamagnetism.

345. Nonelectrolyte electrolyte $\cdot \cdot$ large covalent molecule; dissolves without ionizing.

346. Ionic solid $\cdot \cdot$ its composition implies a very high melting point ($2+/2-$ ions), brittleness, and conduction only when molten.

347. $8.3 \times 10^{-5} \text{ m}$

348. Fluorine

349. $\text{D} \rightarrow \text{A}$ $\cdot \cdot$ this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.

350. 2s, 3p, 4s, 3d, 4d

351. $\text{HCl}(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{Cl}^-(aq) + \text{H}_3\text{O}^+(aq)$

352. Colloidal particles carry like surface charges that repel one another and stay dispersed; the added ions neutralize those charges, removing the repulsion so the particles aggregate and settle out.

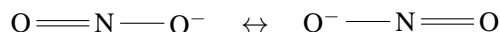
353. $4.1 \times 10^0 \text{ mg}$

354. honey > motor oil > H₂O > CH₃COCH₃ ∴ viscosity ∝ IMF strength and molecular complexity; honey: complex mix of sugars; extensive H-bonding; motor oil: long polymer-like chains; high molecular mass; strong LDF; H₂O: H-bonding; small molecule; CH₃COCH₃: dipole-dipole; small molecule.
355. 5631.2 g
356. $i = \frac{\Delta T}{K_f m} = \frac{0.0930}{1.86 \times 0.0500} = 1.00$; the solute is a nonelectrolyte that stays as whole molecules, so $i = 1$ as expected.
357. CH₃OH + CO → CH₃COOH

+

Li

- 358.
359. 900 zJ ∴ "C" is the level with the third-lowest energy, AKA the second excited state.
360. 4
361. dichlorine heptoxide
362. Consistent. ∴ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which matches the molar mass of HBr.
363. 8.17 mM
364. 4.95 atm. ∴ no gas is added or removed before equilibrium is reached, the amount of gas remains constant. The initial pressure equals the factor by which the gas expands when the final pressure is atmospheric pressure.
365. 9.82×10^4 Pa ∴ the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
366. 5.8795×10^2 nm
367. Solution A: $\Delta T = i K m = 2 \times 0.512 \times 0.500 = 0.512$ °C; Solution B: $\Delta T = 1 \times 0.512 \times 0.500 = 0.256$ °C. Solution A has the greater boiling point elevation ∴ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).
368. Extensive property. Surface area is an extensive property ∴ larger samples have greater total surface area.
369. Water flows from pure water into the sucrose solution. ∴ pure water has zero osmotic pressure; water moves toward higher solute concentration.
370. Ag₂S + 2 H₃OCN → (H₃O)₂S + 2 AgCN
371. $0.2 \frac{\text{g}}{\text{mL}}$
372. Vapor pressure increases ∴ a larger fraction of molecules gains enough kinetic energy to escape the liquid, so more enter the vapor before equilibrium is re-established.
373. $U = \Delta H_f - \Delta H_{\text{sub}} - IE_1 - \frac{1}{2} D - EA = -394 - 89 - 419 - 112 - (-325) = -689$ kJ/mol
374. Heterogeneous mixture ∴ it is a combination of two or more distinct substances that are not uniformly distributed.
375. 2 H₂O (l) + Ba (s) → Ba(OH)₂ (aq) + H₂ (g)
- 376.



the bond order of each N–O bond is 1.5 ∴ the bond order of a bond in a resonance hybrid is the average of its bond orders in the equivalent resonance structures; all of the resonance structures contribute equally, so every N–O bond is identical in the resonance hybrid. Sum the bond order of each N–O bond in NO₂[−] and divide by the number of equivalent resonance structures.

377. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +717 + (-681) = +17.2$ kJ/mol; endothermic ∴ $\Delta H_{\text{sol}} > 0$.
378. 263 K
379. $a = 2\sqrt{2}r = 2\sqrt{2} \times 143 = 404.5$ pm
380. $q = n\Delta H_{\text{fus}} = \frac{13.6 \text{ g}}{46.068 \text{ g/mol}} \times 4.6 \text{ kJ/mol} = 1.36$ kJ
381. $x = 6$
382. 7.32×10^5 Pa. ∴ rearranging the combined gas law gives $P_1 = P_2(V_2/V_1)(T_1/T_2)$. Pressure is directly proportional to absolute temperature and inversely proportional to volume for a fixed amount of gas, so both changes must be accounted for simultaneously.

383. $5 \frac{\text{mol}}{\text{L}}$
384. 180 zJ ∴ "B" is the level with the second-lowest energy, AKA the first excited state.
385. ABAB, repeating ∴ in hexagonal closest packing (HCP) the third layer sits directly above the first layer, distinguishing it from the other close-packed arrangement.
386. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
387. Freezing all 59.7 g would release 19.9 kJ , but only 6.24 kJ is removed ∴ partial freeze: an ice–water mixture at 0°C , with 18.7 g frozen.
388. $1.66 \times 10^{-6} \text{ m}$ ∴ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$
389. $1.90 \times 10^4 \text{ Pa}$ ∴ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
390. 36.5 g/mol
391. Homogeneous mixture ∴ it is a combination of two or more substances with a uniform composition and appearance.
392. $m = \frac{\Delta T_f}{K_f} = \frac{0.186}{1.86} = 0.100 \text{ mol/kg}$; $n = m \times m_{\text{solvent}} = 0.100 \times 0.200 \text{ kg} = 0.0200 \text{ mol}$; $\mathcal{M} = \frac{1.56 \text{ g}}{0.0200 \text{ mol}} = 78.1 \text{ g/mol}$
393. $7 \frac{\text{mol}}{\text{L}}$
394. 24 valence electrons ∴ sum the group valence electrons of every atom; 0 lone pairs on the central B atom
395. 0.2204 mL of barium iodide
396. Covalent-network solid (SiO_2) ∴ covalent-network (quartz): Si–O bonds extend throughout solid.
397. $9.6 \times 10^4 \text{ Pa}$ ∴ the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.
398. 75 zJ ∴ "B" is the level with the second-lowest energy, AKA the first excited state.
399. Basis: 100 g solution $\rightarrow 25.0 \text{ g KCl}$, 75.0 g water; $n_{\text{solute}} = \frac{25.0 \text{ g}}{74.55 \text{ g/mol}} = 0.335 \text{ mol}$; $m = \frac{0.335 \text{ mol}}{0.0750 \text{ kg}} = 4.47 \text{ mol/kg}$
400. $i = \frac{\Delta T}{K_f m} = \frac{0.186}{1.86 \times 0.0500} = 2.00$; the solute is dissociating completely into 2 ions, matching the ideal $i = 2$.
401. $\text{CH}_4\text{N}_2\text{O}$ ($i = 1$) < MgBr_2 ($i = 3$) < Na_2SO_4 ($i = 3$) ∴ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. $\text{CH}_4\text{N}_2\text{O}$: nonelectrolyte; 1 particle; MgBr_2 : strong electrolyte; $\text{Mg}^{2+} + 2\text{Br}^- \rightarrow 3$ particles; Na_2SO_4 : strong electrolyte; $2\text{Na}^+ + \text{SO}_4^{2-} \rightarrow 3$ particles.
402. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \pi_{2p}^2 \pi_{2p}^{*2} \pi_{2p}^{*2}$
Bond order = $\frac{8-6}{2} = 1$
 O_2^{2-} (peroxide) is diamagnetic. The bond order is 1 because there are 8 bonding and 6 antibonding electrons. It is diamagnetic because all electrons are paired.
403. AlF_3 ∴ smaller ion(s) $\rightarrow$ stronger electrostatic attraction; see Coulomb's law ($F \propto \frac{q_1 q_2}{r^2}$)
404. CaI_2
405. Yes, aluminum and oxygen form an ionic compound with the empirical formula Al_2O_3 .
406. 12%
407. Sulfurous acid
408. The experimentally observed bond angle(s) are inconsistent with the orientations of unmodified atomic orbitals ∴ hybridization mixes the O atom's s and p orbitals to produce sp^2 hybrid orbitals whose orientations are consistent with the observed geometry.
409. e^- -group arrangement: Trigonal bipyramidal
Shape: Linear
Ideal bond angle(s): 180°
410. $\text{B}_2 = \text{F}_2$ ($\text{BO}=1$) < C_2 ($\text{BO}=2$). Individual bond orders: B_2 : $\text{BO}=1$, C_2 : $\text{BO}=2$, F_2 : $\text{BO}=1$. ∴ bond order depends on the balance between electrons in bonding and antibonding molecular orbitals: electrons added to bonding orbitals increase bond order, whereas electrons added to antibonding orbitals decrease it. A complete justification correctly accounts for the MO occupancy of each species and uses it to explain the ranking, not merely state the bond-order values.
411. The claim is incorrect. Bond type depends on the orientation of orbital overlap, not on the orbital type alone. ∴ two p orbitals overlapping end-to-end form a σ bond, whereas two p orbitals overlapping side-by-side form a π bond.

412. 19%



413. NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.

414. Hg (485.5 mN/m) > H₂O (72.8 mN/m) > C₆H₁₄ (18.4 mN/m) > CH₄ (13.0 mN/m) ∴ surface tension ∝ IMF strength; Hg: metallic bonding; extremely strong cohesion; H₂O: extensive hydrogen bonding; C₆H₁₄: London dispersion forces only; CH₄: very weak dispersion forces.

415. X is a pure substance; Mass of E₁: 6.9 g. X is a pure substance ∴ the ratio of the masses of E₁ to E₂ is the same for all four samples. A pure substance always has a constant composition. If $m_{E1} + m_{E2} = 10.0\text{ g}$, and you have calculated the mass fraction $m_{E1}/(m_{E1} + m_{E2})$ from the samples, you would expect m_{E1} to be 6.9 g.

416. 8.5 mL of aluminum chlorite

417. the C–N bond in CH₃NH₂. ∴ the Lewis structures, including any resonance contribution where applicable, must first be used to determine the effective amount of bonding electron density shared between the nuclei. Smaller bonding electron density weakens the attractive interaction and places the minimum of the potential-energy curve at a larger internuclear separation, producing the longer equilibrium bond.

418. HBr. ∴ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 80.91 g/mol. Only HBr has that molar mass, so it is the only identity consistent with the measurements.

419. $0.16 \frac{\text{g}}{\text{mL}}$

420. $q = n\Delta H_{\text{fus}} = \frac{7.82\text{ g}}{78.114\text{ g/mol}} \times 9.87\text{ kJ/mol} = 0.988\text{ kJ}$

421. Pico (symbol: p)

422. $7.398 \times 10^0\text{ g}$

423. NH₂O

424. Arsenic acid

425. It is a heterogeneous mixture. ∴ it is a combination of two or more distinct substances that are not uniformly distributed.

426. $9.23 \times 10^1\text{ }\mu\text{g}$

427. Sulfur

428. H₃O₂Br

429. $4\text{ O}_2 + 3\text{ H}_2 + \text{N}_2 \rightarrow 2\text{ H}_2\text{O} + 2\text{ HNO}_3$

430. paramagnetic ∴ the MO configuration contains 1 unpaired electron.

431. Liquid

432. $0.45 \frac{\text{g}}{\text{mL}}$

433. $3.73 \times 10^{-7}\text{ m}$ ∴ $\Delta E = \left(\frac{-R_y}{n_2^2}\right) - \left(\frac{-R_y}{n_1^2}\right)$ and $\Delta E = \frac{hc}{\lambda}$

434. 20.77 g

435. The solid is denser ∴ a positive slope means $\Delta V_{\text{fus}} < 0$, i.e. the solid is more dense than the liquid.

436. 223.22 mL

437. 81.91%

438. 6.07 g

439. polar covalent ($\Delta\text{EN} = 1.43$); F carries the δ^- ∴ $0.4 \leq \Delta\text{EN} < 2.0$ and F is the more electronegative atom

440. Water flows out of the cell; the cell loses water and plasmolyzes. ∴ the solution is hypertonic: particle concentration = $2 \times 0.200\text{ M} = 0.400\text{ osmol/L} > 0.300\text{ osmol/L}$ inside.

441. $3.0 \times 10^{-6}\text{ moles}$

442. 87%

443. Copper

444. $K_H = C/P = 1.3 \times 10^{-3}/1.0\text{ atm} = 1.3 \times 10^{-3}\text{ mol/(L}\cdot\text{atm)}$; $P = \frac{C}{K_H} = \frac{5.2 \times 10^{-3}\text{ mol/L}}{1.3 \times 10^{-3}\text{ mol/(L}\cdot\text{atm)}} = 4.0\text{ atm}$

445. Molality (or mole fraction). $\therefore$ colligative formulas need a temperature-independent count of particles per solvent; molarity would drift as temperature changes the solution volume.
446. A σ bond forms $\therefore$ direct head-on overlap concentrates electron density between the nuclei along the internuclear axis, whereas π bonds require lateral overlap of parallel orbitals.
447. The heat breaks intermolecular forces instead of raising kinetic energy $\therefore$ during a phase change energy is stored as intermolecular potential energy as solid becomes liquid, so average kinetic energy (temperature) is unchanged.
448. 1.00 mL. $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{H}_2:\text{N}_2$ is 3:1 so the volume changes by the same factor.
449. 2.3 moles
450. 4.2 L. $\therefore$ the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product PV remains unchanged.
451. 1.2×10^3 mol
452. A_2B_4
453. 6.9 mL
454. Cation:anion = 1 : 1 $\therefore$ counting the ions per unit cell from a simple cubic array of anions with the cation in the cubic (body-center) hole gives that ratio (the cesium chloride (CsCl) structure).
455. Both students must assign the same hybridization: sp^2 . The hybridization is determined by the number of electron domains around the central atom (3), which fixes the σ framework. Moving a double bond changes only the distribution of π electrons and formal charges, so the electron-domain geometry and hybridization remain unchanged.
456. P_2Se_3 is a molecular compound.
457. 79%
458. 3.0×10^{-11} g
459. 16.5 g
460. Gel; e.g., gelatin. $\therefore$ a liquid-in-solid dispersion defines this colloid type.
461. $2\text{H}_3\text{BO}_3 + 3\text{Zn}(\text{OH})_2 \rightarrow 6\text{H}_2\text{O} + \text{Zn}_3(\text{BO}_3)_2$
462. 1.2 atm. $\therefore$ pressure depends on the total amount of gas particles in the container, the amounts of both gases are added before applying the ideal gas law.
463. $m = \frac{0.0438 \text{ mol}}{0.250 \text{ kg}} = 0.175 \text{ mol/kg}$; $\Delta T_b = iK_b m = 1 \times 0.512 \times 0.175 = 0.0897^\circ\text{C}$; $T_b = 100.0 + 0.0897 = 100.^\circ\text{C}$
464. sp^3d^2 ; octahedral electron-group geometry; octahedral molecular geometry; 90° $\therefore$ the Lewis structure contains 6 electron groups around the central atom. 6 electron groups require an octahedral arrangement, so the central atom uses sp^3d^2 hybrid orbitals to orient those electron groups for minimum repulsion. The arrangement of bonding pairs and lone pairs within that electron-group geometry produces the observed molecular geometry and corresponding bond angle(s).
465. 288 kJ
466. No. $\therefore$ within a given hybridization scheme the hybrid orbitals are equivalent because they arise from the same mathematical combination of the original orbitals; they differ only in spatial orientation.
467. It will rise $\therefore$ adhesion to glass exceeds cohesion, pulling the liquid up the tube.
468. The molecular orbital with less electron density between the nuclei has the higher energy $\therefore$ electron density located between the nuclei is attracted to both nuclei simultaneously, increasing the stabilizing electrostatic attraction. Greater density between the nuclei therefore lowers the total energy, whereas less density leaves the nuclei less effectively bound and the energy higher.
469. C_2 is diamagnetic and weakly repelled $\therefore$ all of its electrons are paired, so the magnetic moments cancel and no net moment remains.
470. 2.8% $\therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
471. molecular
472. 8 molecular orbitals $\therefore$ eight atomic orbitals participate, so eight molecular orbitals are formed.
473. 410 g
474. $F = 1 - 1 + 2 = 2$. $\therefore$ Both temperature and pressure may be varied independently because a single phase imposes no phase-equilibrium constraint between them.

475. Increases. $\therefore$ at the same temperature and pressure every ideal gas has the same molar concentration, but each mole of the replacement gas has a larger mass. More mass therefore occupies the same volume.
476. AlF_3 $\therefore$ smaller ion(s) $\rightarrow$ stronger electrostatic attraction; see Coulomb's law ($F \propto \frac{q_1 q_2}{r^2}$)
477. $\Delta H_{\text{vap}} = \frac{-R \ln(P_2/P_1)}{1/T_2 - 1/T_1} = 37.7 \text{ kJ/mol}$
478. $9.86 \times 10^4 \text{ Pa}$ $\therefore$ the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.
479. (lowest energy) red light < yellow light < ultraviolet light < x-rays (highest energy)
480. $6.20 \times 10^{-3} \text{ L}$
481. (shortest wavelength) gamma rays < red light < infrared light < microwaves (longest wavelength)
482. $4.2641 \times 10^3 \text{ g}$
483. $4.790 \times 10^0 \text{ kg}$
484. 280 mL
485. +1 $\therefore FC = \text{valence } e^- - \text{unshared } e^- - \frac{1}{2} \text{shared } e^- \therefore FC = 6 - 2 - \frac{1}{2}(6)$
486. σ/π bonding. Rotational freedom depends on the type of orbital overlap present in the bond. A π bond requires side-by-side overlap that is lost upon rotation, whereas a σ bond retains end-to-end overlap.
487. (lowest energy) yellow light < blue light < ultraviolet light < x-rays (highest energy)
488. $11 \frac{\text{mol}}{\text{L}}$
489. 5 σ bonds and 1 π bond. The π bond comes from the C=C bond. $\therefore$ Counting the σ bond in every covalent bond and the additional π bond(s) present only in the multiple bond(s) gives these totals.
490. $8.02 \times 10^0 \text{ g}$
491. $\text{B}_2 = \text{F}_2$ (BO=1) < C_2 (BO=2). Individual bond orders: B_2 : BO=1, C_2 : BO=2, F_2 : BO=1. $\therefore$ bond order depends on the balance between electrons in bonding and antibonding molecular orbitals: electrons added to bonding orbitals increase bond order, whereas electrons added to antibonding orbitals decrease it. A complete justification correctly accounts for the MO occupancy of each species and uses it to explain the ranking, not merely state the bond-order values.
492. The potential energy increases sharply $\therefore$ forcing the atoms closer together causes repulsive interactions to increase rapidly and dominate over the attractive interactions.
493. $6.77 \times 10^{-3} \text{ mol}$
494. reduced
495. Metallic solid (Au) $\therefore$ delocalized electrons (sea of electrons) allow electrical conductivity and confer malleability.
496. $G \rightarrow F$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
497. $m = \frac{\Delta T}{K_b \cdot 0.512} = \frac{0.186}{0.512} = 0.363 \text{ mol/kg}$; $n = 0.363 \times 0.100 \text{ kg} = 0.0363 \text{ mol}$; $m_{\text{naph}} = 0.0363 \text{ mol} \times 128.17 \text{ g/mol} = 4.66 \text{ g}$
498. 435 kg
499. A σ bond $\therefore$ head-on overlap concentrates electron density along the internuclear axis.
500. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^2 \pi_{2p}^2$
Bond order = $\frac{8-2}{2} = 3$
 C_2 is diamagnetic. The bond order is 3 because there are 8 bonding and 2 antibonding electrons. It is diamagnetic because all electrons are paired.
501. $\Delta H_{\text{vap}} = \frac{-R \ln(P_2/P_1)}{1/T_2 - 1/T_1} = 30.5 \text{ kJ/mol}$
502. 45 zJ $\therefore$ the energy of the absorbed photon is equal to the difference in energy levels. $|90 - 135| = 45$
503. Silver dichromate: $\text{Ag}_2\text{Cr}_2\text{O}_7$
504. $7.66 \times 10^1 \text{ Gm}$
505. Oxygen most commonly forms the anion O^{2-} .
506. e^- -group arrangement: Trigonal planar
Shape: Trigonal planar

Ideal bond angle(s): 120°

507. $P_2 = P_1 \exp \left[-\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right] = 0.476 \text{ atm} \because T_2 = 353.15 \text{ K}, T_1 = 373.15 \text{ K}, R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.
508. Yes $\because$ the de Broglie wavelength $\lambda = \frac{h}{mv}$ is larger than the width of the object.
509. $m = \frac{\Delta T_f}{K_f} = \frac{0.512}{37.7} = 0.0136 \text{ mol/kg}; n = m \times m_{\text{solvent}} = 0.0136 \times 0.100 \text{ kg} = 1.36 \times 10^{-3} \text{ mol}; \mathcal{M} = \frac{0.125 \text{ g}}{1.36 \times 10^{-3} \text{ mol}} = 92.1 \text{ g/mol}$
510. 5844.9 m
511. $6.937 \times 10^1 \text{ g}$
512. $q = n\Delta H_{\text{vap}} = \frac{41.9 \text{ g}}{32.042 \text{ g/mol}} \times 35.2 \text{ kJ/mol} = 46.0 \text{ kJ}$
513. $\rho = \frac{Z \cdot M}{N_A \cdot a^3} = \frac{4 \times 78.075}{6.022 \times 10^{23} \times (546.3 \times 10^{-10})^3} = 3.18 \text{ g/mL}$
514. $2.9\% \because \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
515. No layers form $\because$ methanol is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.
516. Sample B $\because \rho_A = 1.2 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 1.6 \frac{\text{g}}{\text{mL}}$
517. 20.1 g
518. $2.5\% \because \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
519. $3.63 \times 10^1 \text{ cm}$
520. 35%
521. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2$; Sr forms Sr^{2+} and is isoelectronic with Kr $\because$ given the neutral Sr atom's electron configuration, it loses 2 electrons to match the noble-gas electron configuration of Kr.
522. 15.9 g
523. $\text{C}-\text{N} < \text{C}=\text{N} < \text{C}\equiv\text{N}$ $\because$ higher bond order means more shared electron pairs between the nuclei, increasing the attraction between the atoms. Stronger attraction requires more energy to break the bond, so bond strength increases as bond order increases.
524. $\Pi = iMRT = 2 \times 0.154 \times 0.08206 \times 298 \text{ K} = 7.54 \text{ atm}$; minimum applied pressure must exceed 7.54 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.
525. $\Pi = iMRT = 1 \times 0.200 \times 0.08206 \times 298 \text{ K} = 4.89 \text{ atm}$
526. $\text{D} \rightarrow \text{E}$ $\because$ this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.
527. It is the equilibrium separation between molecules $\because$ at the minimum the net force is zero (attraction balances repulsion), and molecules in a solid vibrate about this distance.
528. -292 kJ (exothermic) $\because \Delta H^\circ \approx \sum BE_{\text{broken}} - \sum BE_{\text{formed}} = 2539 - 2831$
529. Solubility increases significantly $\because$ dissolution is strongly endothermic; Le Chatelier: higher T shifts equilibrium toward products.
530. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
531. C: sp — justified by 2 C=O bonds
O: sp^2 — justified by 1 C=O bond and 2 lone pairs
532. $8.0 \times 10^2 \text{ mol}$
533. $K_H = C/P = 6.1 \times 10^{-4} / 1.0 \text{ atm} = 6.1 \times 10^{-4} \text{ mol/(L}\cdot\text{atm)}$; $P = \frac{C}{K_H} = \frac{1.8 \times 10^{-3} \text{ mol/L}}{6.1 \times 10^{-4} \text{ mol/(L}\cdot\text{atm)}} = 3.0 \text{ atm}$
534. $\text{G} \rightarrow \text{A}$ $\because$ this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.
535. $\text{Fe(OH)}_3 + 3\text{HCl} \rightarrow 3\text{H}_2\text{O} + \text{FeCl}_3$
536. $5.1 \times 10^3 \text{ Pa}$. $\because$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
537. X is a mixture $\because$ the ratio of the masses of E_1 to E_2 is not the same in all four samples. Mixtures do not have a fixed composition, so different samples can contain different proportions of the two elements.

538. 38%

539. $\Pi = iMRT = 2 \times 0.600 \times 0.08206 \times 298 \text{ K} = 29.4 \text{ atm}$; minimum applied pressure must exceed 29.4 atm $\therefore$ reverse osmosis requires $P_{\text{applied}} > \Pi$.

540. neither oxidized nor reduced

541. Hydrogen bonding $\therefore$ both molecules have O–H groups that donate and accept hydrogen bonds.

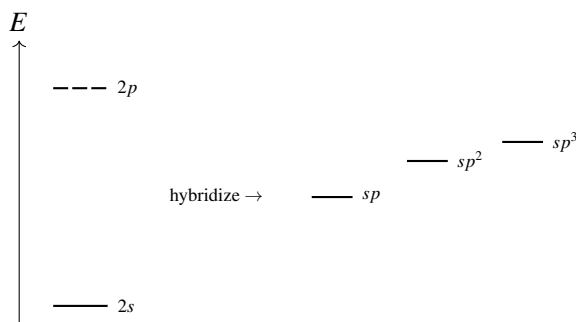
542. $3.984 \times 10^{26} \text{ g}$

543. $n_{\text{met}} = \frac{18.4 \text{ g}}{32.04} = 0.575 \text{ mol}$; $n_{\text{H}_2\text{O}} = \frac{100. \text{ g}}{18.015} = 5.55 \text{ mol}$; $\chi_{\text{met}} = 0.0938$; $\chi_{\text{H}_2\text{O}} = 0.906$; $\chi_{\text{met}} + \chi_{\text{H}_2\text{O}} = 1.00 \checkmark$

544. 5.25 cm

545. $m = \frac{0.0584 \text{ mol}}{0.200 \text{ kg}} = 0.292 \text{ mol/kg}$; $\Delta T_f = iK_f m = 1 \times 1.86 \times 0.292 = 0.543^\circ \text{C}$; $T_f = 0.00 - 0.543 = -0.543^\circ \text{C}$

546. sp^3 $\therefore$ it contains the greater fraction of p character. Because the $2p$ orbitals are higher in energy than the $2s$ orbital, increasing the contribution of $2p$ orbitals raises the weighted-average expectation value, whereas increasing s character lowers it.



547. Ba is a neutral atom. It has 56 protons and 56 electrons.

548. 3 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.

549. $3(\text{H}_3\text{O})_2\text{S} + 2\text{AlCl}_3 \rightarrow 6\text{H}_3\text{OCl} + \text{Al}_2\text{S}_3$

550. Per 1 L of solution: mass solution = $1.02 \times 10^3 \text{ g}$; mass solute = $0.500 \text{ mol} \times 36.46 \text{ g/mol} = 18.2 \text{ g}$; mass solvent = $1.02 \times 10^3 \text{ g} - 18.2 \text{ g} = 997 \text{ g}$; $m = \frac{0.500 \text{ mol}}{0.997 \text{ kg}} = 0.502 \text{ mol/kg}$

551. +3

552. He_2^+ : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*1}$; bond order = $\frac{2-1}{2} = 0.5$; paramagnetic.

He_2 : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*2}$; bond order = $\frac{2-2}{2} = 0$; diamagnetic (not stable).

Removing one electron changes the electron occupancy of the highest-energy MO. Adding an electron to an antibonding orbital *destabilizes* the molecule, *lowering* the bond order, while removing one from an antibonding orbital *stabilizes* the molecule, raising the bond order. An unpaired electron makes the species paramagnetic; if all electrons are paired, it is diamagnetic.

553. $\lambda = \frac{2d \sin \theta}{n} = \frac{2 \times 394 \times \sin(16.9^\circ)}{1} = 229 \text{ pm}$. $\therefore$ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d \sin \theta$, leading to $n\lambda = 2d \sin \theta$.

554. $7.628 \times 10^3 \frac{\text{kg}}{\text{m}^3}$

555. $1.3 \times 10^4 \frac{\text{kg}}{\text{m}^3}$

556. He_2^+ : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*1}$; bond order = $\frac{2-1}{2} = 0.5$; paramagnetic.

He_2 : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*2}$; bond order = $\frac{2-2}{2} = 0$; diamagnetic (not stable).

Removing one electron changes the electron occupancy of the highest-energy MO. Adding an electron to an antibonding orbital *destabilizes* the molecule, *lowering* the bond order, while removing one from an antibonding orbital *stabilizes* the molecule, raising the bond order. An unpaired electron makes the species paramagnetic; if all electrons are paired, it is diamagnetic.

557. $\text{Sr}^{2+}(\text{aq}) + 2\text{F}^-(\text{aq}) \rightarrow \text{SrF}_2(\text{s})$

558. -802 kJ (exothermic) $\therefore \Delta H^\circ \approx \sum BE_{\text{broken}} - \sum BE_{\text{formed}} = 2652 - 3454$

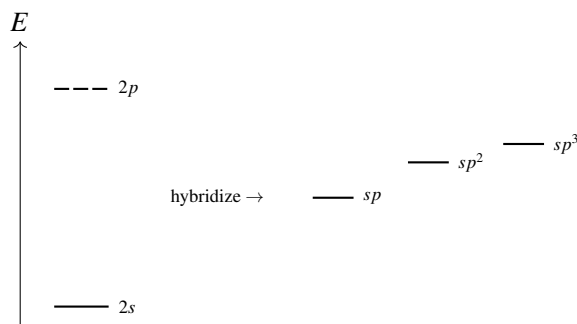
559. $\text{HI}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightarrow \text{I}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$

560. 308 K

561. $0.716 \frac{\text{mol}}{\text{L}}$
562. Ammonia has the higher boiling point. $\therefore$ its hydrogen atoms are bonded directly to nitrogen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.
563. ionic, strong base
564. 1.5 L. $\therefore$ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until $P_i V_i = P_f V_f$. Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
565. $n_{\text{solute}} = \frac{10.0}{342.3} = 0.0292 \text{ mol}$; $n_{\text{water}} = \frac{500.0}{18.015} = 27.8 \text{ mol}$. $\chi_{\text{solute}} = \frac{0.0292}{0.0292 + 27.8} = 1.05 \times 10^{-3}$
566. 0.01880 cm
567. 3.52 g
568. $\text{K}(s) + \text{Na}^+(aq) \rightarrow \text{K}^+(aq) + \text{Na}(s)$
569. 81 kJ
570. 3.24 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 2:1 mole ratio, so the gas volumes must change in the same 2:1 ratio.
571. MO configuration: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^1 \pi_{2p}^1$
 Bond order = $\frac{6-4}{2} = 1$ $\therefore$ bond order is determined by the balance between electrons in bonding (6) and antibonding (4) molecular orbitals. Bonding orbitals stabilize the molecule, whereas antibonding orbitals destabilize it, so a larger excess of bonding electrons produces a stronger bond.
 B_2 is paramagnetic $\therefore$ its magnetic character is determined by whether any molecular orbitals contain unpaired electrons. One or more unpaired electrons produces paramagnetism, whereas complete electron pairing produces diamagnetism.
572. Oxygen
573. Emitted $\therefore$ a photon is emitted when an electron transitions to a lower energy level
574. Solubility increases by a factor of 2 $\therefore C = K_H P$; $C_1 = 6.8 \times 10^{-2} \text{ mol/L}$, $C_2 = 1.4 \times 10^{-1} \text{ mol/L}$; $C_2/C_1 = 4.0/2.0 = 2$.
575. 19 mL
576. 4.7 mm
577. $2\text{HF} + \text{Pb}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{PbF}_2$
578. No, oxygen and bromine cannot form an ionic compound.
579. 14.5 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the $\text{CH}_4:\text{CO}_2$ coefficient ratio of 1:1 determines the required volume.
580. ABCABC, repeating $\therefore$ in cubic closest packing (CCP) the third layer occupies new positions before the pattern repeats, distinguishing it from the other close-packed arrangement.
581. 6.353 g
582. $\text{G} \rightarrow \text{F}$ $\therefore$ because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).
583. 16 valence electrons $\therefore$ sum the group valence electrons of every atom; 0 lone pairs on the central C atom
584. 0 $\therefore$ only unhybridized p orbitals can form π bonds, and 0 unhybridized p orbitals remain after hybridization.
585. $\text{G} \rightarrow \text{A}$ $\therefore$ this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.
586. KNO_3 crystallizes from solution on cooling. $\therefore$ KNO_3 solubility increases with T; cooling creates supersaturation $\rightarrow$ crystallization.
587. 0.0586 g
588. 0.330 moles
589. $\text{C}_2\text{H}_5\text{OH}$
590. ΔH_{vap} is much larger than ΔH_{fus} $\therefore$ plateau length is proportional to heat absorbed; vaporization separates molecules completely, while melting only loosens the lattice.
591. $9.5169 \times 10^3 \text{ m}$
592. Pure substances: B $\therefore$ the mass ratio of E_1 to E_2 in B is approximately an integer multiple of the ratio in pure substance X. When

two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).

593. $m = \frac{q}{\Delta H} = \frac{6.19 \times 10^3 \text{ J}}{247 \text{ J/g}} = 25.1 \text{ g}$
594. 1.82 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
595. $6.49 \times 10^4 \text{ Pa}$ $\therefore$ the mole fractions must sum to 1, so the mole fraction of N_2 is the remainder. Dalton's law states that a gas's partial pressure equals its mole fraction multiplied by the total pressure.
596. 660 mol
597. Double-Displacement
598. $\text{Al}_2\text{Cr}_6\text{O}_21$
599. 285000000000 Hz
600. 0.079 m. $\therefore$ Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through $A = F/P$. Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
601. X-rays are required $\therefore$ diffraction needs a wavelength comparable to the plane spacing ($\sim 100 \text{ pm}$); X-ray wavelengths match atomic spacings, while visible light ($\sim 500 \text{ nm}$) is far too long to resolve them.
602. It is a σ bond $\therefore$ head-on overlap is cylindrically symmetric around the internuclear axis, so rotation does not reduce overlap, whereas rotating a π bond destroys the required side-by-side overlap.
603. $C = K_{\text{H}}P = 3.4 \times 10^{-2} \times 0.25 \text{ atm} = 8.5 \times 10^{-3} \text{ mol/L}$
604. 3.31 m
605. $\text{D} \rightarrow \text{E}$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.
606. NaCl : $i \times M = 2 \times 0.150 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic); Na_2SO_4 : $i \times M = 3 \times 0.150 \text{ M} = 0.450 \text{ osmol/L}$ (hypertonic); MgCl_2 : $i \times M = 3 \times 0.100 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic). $\therefore$ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.
607. Silver sulfide decahydrate: $\text{Ag}_2\text{S} \cdot 10\text{H}_2\text{O}$
608. $T_2 = \left(\frac{1}{T_1} - \frac{R \ln(P_2/P_1)}{\Delta H_{\text{vap}}} \right)^{-1} = 380.6 \text{ K} = 107^\circ\text{C}$ $\therefore$ boiling occurs when the vapor pressure equals the external pressure, and a higher external pressure raises the boiling point.
609. 7.231×10^{21} phosphorus atoms
610. $\chi_{\text{H}_2\text{O}} = \frac{5.55}{5.55 + 0.200} = 0.965$; $P = 0.965 \times 149.4 \text{ mmHg} = 144 \text{ mmHg}$; $\Delta P = 5.19 \text{ mmHg}$ lowering.
611. Hydrogen fluoride has the higher boiling point. $\therefore$ its hydrogen atoms are bonded directly to fluorine, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.
612. $2.17 \times 10^1 \text{ cm}$
613. Mg $\therefore$ effective nuclear charge increases across a period, increasing an atom's attraction for bonding electrons
614. 6.4 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 3:6 particle ratio between Cl_2 and HCl , so their volumes must be in the same ratio.
615. 0.11 m. $\therefore$ Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through $A = F/P$. Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
616. $\chi_{\text{H}_2\text{O}} = \frac{11.1}{11.1 + 0.569} = 0.951$; $P = 0.951 \times 355.1 \text{ mmHg} = 338 \text{ mmHg}$; $\Delta P = 17.3 \text{ mmHg}$ lowering.
617. Emitted $\therefore$ a photon is emitted when an electron transitions to a lower energy level
618. 2.07 g
619. sp^3 $\therefore$ it contains the greater fraction of p character. Because the $2p$ orbitals are higher in energy than the $2s$ orbital, increasing the contribution of $2p$ orbitals raises the weighted-average expectation value, whereas increasing s character lowers it.



620. $7.43 \times 10^1 \mu\text{g}$
621. Both conduct electricity within the carbon sheets. $\therefore$ each consists of hexagonal sheets in which every carbon forms three covalent bonds, leaving electrons delocalized across the sheet that are free to move and carry charge.
622. 8 σ bonds and 1 π bond. The π bond comes from the C=C bond. $\therefore$ Counting the σ bond in every covalent bond and the additional π bond(s) present only in the multiple bond(s) gives these totals.
623. $m = \frac{0.0438 \text{ mol}}{0.250 \text{ kg}} = 0.175 \text{ mol/kg}$; $\Delta T_b = i K_b m = 1 \times 0.512 \times 0.175 = 0.0897^\circ\text{C}$; $T_b = 100.0 + 0.0897 = 100.^\circ\text{C}$
624. $-7.83 \times 10^4 \text{ J}$
625. Ba, Sr, Ca, Mg $\therefore$ elements with greater metallic character hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
626. $2\text{HNO}_3 + \text{Zn}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{Zn}(\text{NO}_3)_2$
627. Sulfur
628. 0.75. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
629. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
630. Calcium oxide: CaO
631. Suspension $\therefore$ particle size is greater than 1000 nm; suspensions are heterogeneous mixtures.
632. 4.12×10^{-49} moles
633. 14 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the $\text{H}_2:\text{Cl}_2$ coefficient ratio of 1:1 determines the required volume.
634. sp hybridization. $\therefore$ The central Be forms two bonds and has no lone pairs, giving two electron groups. Two electron groups adopt a linear electron-domain arrangement, requiring sp hybridization and producing a linear molecular geometry with a 180° bond angle. $\therefore$ the molecular geometry is linear with an ideal bond angle of 180° .
635. 776.48 nm
636. LiCl $\therefore$ greatest boiling point elevation $\therefore$ largest effective molality. CH_3OH : $i \times m = 1 \times \frac{1.00}{32.04} = 0.0312 \text{ mol/kg}$; MgCl_2 : $i \times m = 3 \times \frac{1.00}{95.21} = 0.0315 \text{ mol/kg}$; LiCl: $i \times m = 2 \times \frac{1.00}{42.39} = 0.0472 \text{ mol/kg}$.
637. Amorphous $\therefore$ its random, disordered structure looks the same in all directions and softens over a range, unlike an ordered crystal, which can be anisotropic and melts sharply.
638. 2s, 2p, 3s, 3p, 5s
639. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +929 + (-904) = +1.9 \text{ kJ/mol}$; endothermic $\therefore \Delta H_{\text{sol}} > 0$.
640. It dissolves caffeine yet penetrates and evaporates cleanly $\therefore$ liquid-like density dissolves caffeine while gas-like low viscosity penetrates the beans; releasing pressure evaporates it, leaving no residue.
641. Hydrogen bonding $\therefore$ O–H bonds with lone pairs on O allow H-bonding.
642. $2\text{H}_2\text{O}(\text{l}) + \text{Ca}(\text{s}) \rightarrow \text{Ca}(\text{OH})_2(\text{aq}) + \text{H}_2(\text{g})$
643. Na $\therefore$ atoms on the left hold their valence electrons less strongly and lose them more easily
644. Exception: electron-deficient $\therefore$ the central Be atom has only 4 electrons (fewer than an octet)
645. Physical change $\therefore$ the colors combine physically without a chemical reaction.

646. $\text{C}_3\text{H}_8\text{O}_3 > \text{C}_2\text{H}_5\text{OH} > \text{H}_2\text{O} > \text{C}_6\text{H}_{14}$ $\therefore$ viscosity $\propto$ IMF strength and molecular complexity; $\text{C}_3\text{H}_8\text{O}_3$: extensive H-bonding network across 3 $-\text{OH}$ groups; $\text{C}_2\text{H}_5\text{OH}$: H-bonding; slightly less than water due to hydrophobic tail; H_2O : strong H-bonding but small molecule; C_6H_{14} : LDF only; low viscosity.
647. London dispersion, dipole–dipole, and hydrogen bonding $\therefore$ polar with N–H bonds, so all three forces act.
648. $x = 6$
649. $\text{C}_2\text{H}_5\text{NH}_2 > \text{C}_3\text{H}_7\text{NH}_2 > \text{C}_6\text{H}_{13}\text{NH}_2 > \text{C}_8\text{H}_{17}\text{NH}_2$ $\therefore$ all share the same H-bonding $-\text{NH}_2$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.
650. $1.5 \times 10^0 \text{ mm}$
651. $\text{B} \rightarrow \text{C}$ $\therefore$ because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).
652. Pure substance. It is a pure substance (element); diamond consists entirely of carbon atoms in a crystal lattice.
653. $m = \frac{\Delta T}{K_f = 1.86} = \frac{0.512}{1.86} = 0.275 \text{ mol/kg}$; $n = 0.275 \times 0.200 \text{ kg} = 0.0551 \text{ mol}$; $m_{\text{anth}} = 0.0551 \text{ mol} \times 178.23 \text{ g/mol} = 9.81 \text{ g}$
654. $6.16 \times 10^{-1} \text{ g}$
655. 333 K
656. 0.09513 moles
657. $6.86 \frac{\text{m}}{\text{s}}$
658. $\text{CH}_3\text{NH}_2 > \text{C}_4\text{H}_9\text{NH}_2 > \text{C}_6\text{H}_{13}\text{NH}_2 > \text{C}_8\text{H}_{17}\text{NH}_2$ $\therefore$ all share the same H-bonding $-\text{NH}_2$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.
659. 2 protons and 2 neutrons.
660. 0.0000000000000069 s
661. 6.7%. $\therefore$ the fertilizer's potassium content is the sum of the potassium supplied by each ingredient, so each ingredient's contribution equals (mass percent of ingredient) $\times$ (mass fraction of potassium in that compound), and these contributions are added together.
662. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
663. $\Pi = iMRT = 2 \times 0.600 \times 0.08206 \times 298 \text{ K} = 29.4 \text{ atm}$; minimum applied pressure must exceed 29.4 atm $\therefore$ reverse osmosis requires $P_{\text{applied}} > \Pi$.
664. Physical property. The color of copper is a physical property; it describes the appearance of copper without any chemical change.
665. $\text{E} \rightarrow \text{A}$ $\therefore$ because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).
666. No. $\therefore$ keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
667. $2.16 \times 10^3 \text{ mol}$. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
668. Yes $\therefore$ London dispersion forces are similar in I_2 and CCl_4 ; like dissolves like.
669. $\rho = \frac{ZM}{N_A a^3} = \frac{4 \times 107.868}{6.022 \times 10^{23} \times (408.53 \times 10^{-10})^3} = 10.5 \text{ g/cm}^3$. $\therefore$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.
670. 0.0000000000000032 s
671. $2.258 \times 10^0 \text{ km}$
672. $\Pi = iMRT = 1 \times 0.0500 \times 0.08206 \times 298 \text{ K} = 1.22 \text{ atm}$
673. $5.4425 \times 10^{-4} \text{ moles}$
674. 1-propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$). $\therefore$ Both molecules are polar, but 1-propanol can also form hydrogen bonds because it contains an O–H bond. Hydrogen bonding strengthens the attractions between molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.
675. Bromine

676. The SI base unit for length is the meter (symbol: m).
677. e^- -group arrangement: Octahedral
Shape: Square planar
Ideal bond angle(s): 90°
678. 0.0107 g
679. Pure substance. It is a pure substance (compound); it has a definite molecular formula and a fixed composition.
680. 72 g/mol. $\therefore$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
681. 4.8×10^{-11} g
682. sp^3 ; tetrahedral electron-group geometry; trigonal pyramidal molecular geometry; $<109.5^\circ$ $\therefore$ the Lewis structure contains 4 electron groups around the central atom. 4 electron groups require a tetrahedral arrangement, so the central atom uses sp^3 hybrid orbitals to orient those electron groups for minimum repulsion. The arrangement of bonding pairs and lone pairs within that electron-group geometry produces the observed molecular geometry and corresponding bond angle(s).
683. 0.0146 L
684. neither oxidized nor reduced
685. 0.122 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
686. 5.75 cm^3 . $\therefore$ at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:1 ratio of H_2 to CH_3OH , so the product volume is multiplied by 1/2.
687. 57.3 fm
688. 3.3×10^{51} g
689. 14 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:4 particle ratio between N_2O_5 and NO_2 , so their volumes must be in the same ratio.
690. $m = \frac{q}{\Delta H} = \frac{23.2 \times 10^3 \text{ J}}{1370 \text{ J/g}} = 16.9 \text{ g}$
691. SiS_2
692. $A \rightarrow G$ $\therefore$ this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.
693. $267.2 \frac{\text{g}}{\text{mol}}$
694. Amorphous $\therefore$ its random, disordered structure looks the same in all directions and softens over a range, unlike an ordered crystal, which can be anisotropic and melts sharply.
695. 7.2%. $\therefore$ the fertilizer's potassium content is the sum of the potassium supplied by each ingredient, so each ingredient's contribution equals (mass percent of ingredient) $\times$ (mass fraction of potassium in that compound), and these contributions are added together.
696. (lowest frequency) radio waves $<$ red light $<$ violet light $<$ ultraviolet light (highest frequency)
697. No $\therefore$ the de Broglie wavelength $\lambda = \frac{h}{mv}$ is smaller than the width of the object.
698. 90 zJ $\therefore$ "C" is the level with the third-lowest energy, AKA the second excited state.
699. 65.31 MPa
700. X is a pure substance; Mass of E_1 : 7.3 g. X is a pure substance $\therefore$ the ratio of the masses of E_1 to E_2 is the same for all four samples. A pure substance always has a constant composition. If $m_{E1} + m_{E2} = 10.0 \text{ g}$, and you have calculated the mass fraction $m_{E1}/(m_{E1} + m_{E2})$ from the samples, you would expect m_{E1} to be 7.3 g.
701. He (760 atm) $>$ N_2 (5.7 atm) $>$ CO_2 (0.2 atm) $>$ H_2O (0.031 atm) $\therefore$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O : strong H-bonding; CO_2 : LDF + weak dipole; N_2 : LDF only; small; He : LDF only; tiny).
702. $\text{He} < \text{Ar} < \text{Kr} < \text{Xe}$ $\therefore$ larger molar mass $\rightarrow$ greater polarizability $\rightarrow$ stronger LDF $\rightarrow$ higher BP. Molar masses: $\text{He} = 4.0 \text{ g/mol}$, $\text{Ar} = 39.95 \text{ g/mol}$, $\text{Kr} = 83.8 \text{ g/mol}$, $\text{Xe} = 131.3 \text{ g/mol}$.
703. Ionic solid (MgO) $\therefore$ ions held by electrostatic forces; ions mobile only when molten.
704. 17.9 mL

705. 0 $\therefore$ 4 regions of electron density require sp^3 hybridization, which incorporates 3 of the three p orbitals into hybrid orbitals, leaving 0 unhybridized.
706. 52.4% $\therefore$ 1 sphere of radius $r = a/2$; sphere vol = $(4/3)\pi r^3$; cell vol = a^3 ; efficiency = $(4/3)\pi(a/2)^3/a^3 = \pi/6 \approx 52.4\%$.
707. $2\text{HF} + \text{Co}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{CoF}_2$
708. $\text{Xe} > \text{Kr} > \text{Ne} > \text{He}$ $\therefore$ larger molar mass $\rightarrow$ greater polarizability $\rightarrow$ stronger LDF $\rightarrow$ higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.
709. -546 kJ . Since $\Delta H < 0$, the reaction is exothermic.
710. $7.55 \times 10^3\text{ Pa}$. $\therefore$ the balanced equation determines the amount of CO_2 produced from the amount of NaHCO_3 that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
711. Octahedral holes, coordination number 6 $\therefore r_+/r_- = 100/140 = 0.71$ lies in the 0.414–0.732 range, which geometrically favors octahedral coordination.
712. No p orbitals remain unhybridized $\therefore$ all three p orbitals are incorporated into the hybrid set.
713. $8.319 \times 10^1\text{ g}$
714. Obeys the octet rule $\therefore$ the central O atom is surrounded by 8 electrons
715. $4\text{Al}(\text{OH})_3 + 3\text{C} \rightarrow 6\text{H}_2\text{O} + 4\text{Al} + 3\text{CO}_2$
716. 66 g
717. No phase transition (the gas passes continuously into a supercritical fluid) $\therefore$ above the critical temperature no liquid–vapor boundary exists to cross, so density simply rises.
718. $\text{E} \rightarrow \text{A}$ $\therefore$ because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).
719. $4.732 \times 10^{51}\text{ g}$
720. e^- -group arrangement: Trigonal bipyramidal
Shape: Trigonal bipyramidal
Ideal bond angle(s): 90° (ax), 120° (eq)
721. It is a π bond $\therefore$ lateral overlap leaves a node along the internuclear axis and places electron density above and below the axis, whereas a σ bond concentrates electron density directly between the nuclei.
722. 2.47 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
723. 70.7%
724. Melting all 25.8 g uses 8.62 kJ; the remaining heat warms the water $\therefore$ liquid water at $T = \frac{q - mH_{\text{fus}}}{mc} = 51.8^\circ\text{C}$.
725. ionic; high melting point $\therefore$ its ions are held by strong electrostatic attractions throughout the lattice
726. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
727. 8.87 g
728. Ethylene glycol $\therefore$ its two O–H groups plus larger size give more extensive intermolecular attraction than water; stronger intermolecular forces increase viscosity.
729. $3.79 \times 10^4\text{ Pa}$ $\therefore$ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
730. $\text{A} \rightarrow \text{B}$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.
731. N-pentane $\therefore$ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.
732. $7.57 \times 10^{-7}\text{ m}$
733. He (760 atm) $>$ N_2 (5.7 atm) $>$ CO_2 (0.2 atm) $>$ H_2O (0.031 atm) $\therefore$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O : strong H-bonding; CO_2 : LDF + weak dipole; N_2 : LDF only; small; He: LDF only; tiny).
734. $8.75 \times 10^{-7}\text{ m}$ $\therefore \Delta E = \left(\frac{-R_y}{n_2^2}\right) - \left(\frac{-R_y}{n_1^2}\right)$ and $\Delta E = \frac{hc}{\lambda}$
735. The following compounds have a dipole moment: H_2S , CO , OF_2 , NO

736. 10 °C
737. 3.007 kg
738. $q = mc\Delta T = 37.8 \times 2.09 \times 12 = 0.948 \text{ kJ}$
739. 0.20 m
740. e⁻-group arrangement: Tetrahedral
Shape: Bent
Ideal bond angle(s): <109.5°
741. $\Delta H_{\text{vap}} = 83.5 \text{ kJ/mol} \therefore -R \frac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)
742. 52.4% $\therefore$ 1 sphere of radius $r = a/2$; sphere vol = $(4/3)\pi r^3$; cell vol = a^3 ; efficiency = $(4/3)\pi(a/2)^3/a^3 = \pi/6 \approx 52.4\%$.
743. 300 zJ $\therefore$ the energy of the absorbed photon is equal to the difference in energy levels. $|700 - 1000| = 300$
744. $K_H = C/P = 1.3 \times 10^{-3} / 1.0 \text{ atm} = 1.3 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{atm})$; $P = \frac{C}{K_H} = \frac{5.2 \times 10^{-3} \text{ mol/L}}{1.3 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{atm})} = 4.0 \text{ atm}$
745. 50.0% $\therefore$ percent ionic character = $\frac{\mu_{\text{meas}}}{\mu_{\text{ionic}}} \times 100$, where $\mu_{\text{ionic}} = e \cdot d = 4.90 \text{ D} \therefore \frac{2.45}{4.90} \times 100$
746. Decane (C₁₀H₂₂). $\therefore$ Water forms hydrogen bonds, but decane's much larger electron cloud and greater surface area produce dispersion forces that are collectively even stronger. More energy is therefore needed to separate decane molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.
747. 8 $\therefore$ center atom touches all 8 corner atoms.
748. 4s: $n = 4, l = 0, N_e = 2 \dots$ 3s: $n = 3, l = 0, N_e = 2 \dots$ 1s: $n = 1, l = 0, N_e = 2$
749. $\Delta H_{\text{vap}} = 32.5 \text{ kJ/mol} \therefore -R \frac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)
750. sp^3 hybridization. $\therefore$ The phosphorus has three bonding groups and one lone pair, giving four electron groups. Four electron groups adopt a tetrahedral electron-domain arrangement requiring sp^3 hybridization; the lone pair compresses the bond angle, producing a trigonal pyramidal molecular geometry with bond angles less than 109.5°. $\therefore$ the molecular geometry is trigonal pyramidal with an ideal bond angle of <109.5°.

•

Fr

- 751.
752. Sample B $\therefore \rho_A = 1.2 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 8.1 \frac{\text{g}}{\text{mL}}$
753. 0.351 M
754. castor oil > HOCH₂CH₂OH > H₂O > CH₃COCH₃ $\therefore$ viscosity $\propto$ IMF strength and molecular complexity; castor oil: large triglycerides with strong dispersion forces; HOCH₂CH₂OH: strong H-bonding from two alcohol groups; H₂O: hydrogen bonding but low molar mass; CH₃COCH₃: small polar molecule with weaker IMF.
755. the N–N bond in N₂H₄. $\therefore$ the Lewis structures, including any resonance contribution where applicable, must first be used to determine the effective amount of bonding electron density shared between the nuclei. Smaller bonding electron density creates a shallower potential-energy well, so less energy is required to separate the atoms completely.
756. MO configuration: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^1$
Bond order = $\frac{3-2}{2} = 0.5 \therefore$ bond order is determined by the balance between electrons in bonding (3) and antibonding (2) molecular orbitals. Bonding orbitals stabilize the molecule, whereas antibonding orbitals destabilize it, so a larger excess of bonding electrons produces a stronger bond.
Li₂⁺ is paramagnetic $\therefore$ its magnetic character is determined by whether any molecular orbitals contain unpaired electrons. One or more unpaired electrons produces paramagnetism, whereas complete electron pairing produces diamagnetism.
757. Colloid $\therefore$ particle size is within the colloidal range (1-1000 nm); colloidal particles remain dispersed.
758. 0.176. $\therefore$ 27.7 g of aluminum corresponds to 1.03 mol. The reaction produces 1.54 mol of hydrogen. The amount of neon remains 0.328 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
759. No, Tyndall effect is not observed. True solution $\therefore$ sucrose molecules < 1 nm; homogeneous at molecular level; no Tyndall effect.

760. 0.3149 moles
761. Intensive property. Density is an intensive property $\therefore$ the ratio of mass to volume does not depend on how much of the substance is present.
762. 100 zJ $\therefore$ "B" is the level with the second-lowest energy, AKA the first excited state.
763. $5.28 \times 10^{-1} \text{ m}$
764. 0.000000000000051 s
765. The experimentally observed bond angle(s) are inconsistent with the orientations of unmodified atomic orbitals $\therefore$ hybridization mixes the Xe atom's s and p orbitals to produce sp^3d hybrid orbitals whose orientations are consistent with the observed geometry.
766. It has liquid-like density but gas-like viscosity $\therefore$ high density supplies frequent solute contacts (dissolving power) while weak intermolecular resistance permits rapid, gas-like diffusion.
767. The liquid is vaporizing; use $q = n\Delta H_{\text{vap}}$ $\therefore$ the added heat converts liquid to gas at constant temperature, raising potential energy by breaking intermolecular forces rather than raising kinetic energy.
768. Kr (-153°C) > Ar (-186°C) > Ne (-246°C) > He (-269°C). $\therefore$ Each atom interacts only through London dispersion forces. Larger electron clouds are more easily distorted, producing stronger instantaneous dipoles and higher boiling points.
769. $9.9 \times 10^4 \text{ Pa}$ $\therefore$ the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
770. $0.175 \frac{\text{mol}}{\text{L}}$
771. 8.8433 g
772. 1.04 moles
773. He < Ne < Ar < Kr $\therefore$ larger molar mass $\rightarrow$ greater polarizability $\rightarrow$ stronger LDF $\rightarrow$ higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Ar = 39.95 g/mol, Kr = 83.8 g/mol.
774. $\Delta H_{\text{fus}} = \Delta H_{\text{sub}} - \Delta H_{\text{vap}} = 62.36 - 43.3 = 19.06 \text{ kJ/mol}$
775. Hybridization: sp^3d . Electron-Group Geometry: trigonal bipyramidal. Molecular Geometry: T-shaped. $\therefore$ there are 3 bonds and 2 lone pairs around the central atom, there are 5 total electron groups surrounding it and $\therefore$ each requires one hybrid orbital, the central atom is sp^3d hybridized. The electron-group geometry is trigonal bipyramidal. 2 lone pairs change the molecular geometry from trigonal bipyramidal to T-shaped and compress the bond angles to $\leq 90^\circ$.
776. $8.19 \times 10^8 \text{ W}$
777. 15 mM
778. I_2 (184°C) > Br_2 (59°C) > Cl_2 (-34°C) > F_2 (-188°C). $\therefore$ All are nonpolar, so London dispersion forces dominate. Increasing electron count produces more polarizable electron clouds, strengthening dispersion forces, requiring more energy to separate the molecules, and therefore increasing the boiling point.
779. $4\text{Al}(\text{OH})_3 + 3\text{C} \rightarrow 6\text{H}_2\text{O} + 4\text{Al} + 3\text{CO}_2$
780. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
781. H_2SeO_4
782. Solid $\therefore$ P is below the triple-point pressure (0.124 atm) and T is below the triple-point temperature (-210°C).
783. 21 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
784. 0.0025 m
785. molecular
786. $\text{E} \rightarrow \text{A}$ $\therefore$ because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).
787. The rock salt (NaCl) structure $\therefore$ a face-centered cubic array of anions with cations in all of the octahedral holes is its defining arrangement.
788. H_2 : σ_{1s}^2 ; unpaired MO(s): none; diamagnetic. H_2^- : $\sigma_{1s}^2\sigma_{1s}^{*1}$; unpaired MO(s): σ_{1s}^* ; paramagnetic. H_2^- is more strongly attracted because it contains more unpaired electrons.
789. $i = 2$; $\Pi = iMRT = 2 \times 0.500 \times 0.08206 \times 298 \text{ K} = 24.5 \text{ atm}$
790. (shortest wavelength) gamma rays < ultraviolet light < yellow light < radio waves (longest wavelength)
791. KF (-821 kJ/mol) < NaF (-923 kJ/mol) < LiF (-1037 kJ/mol) $\therefore$ lattice energy becomes more negative with higher ion charges

and smaller ionic radii (Coulomb's law: $U \propto \frac{q+q-}{r}$).

792. C_9H_{16}

793. 0.0040 m

794. $q = m\Delta H_{\text{vap}} = 33.1 \times 2260 = 74.8 \text{ kJ}$

795. MgCl_2 : $i \times M = 3 \times 0.100 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic); Na_2SO_4 : $i \times M = 3 \times 0.100 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic); CaCl_2 : $i \times M = 3 \times 0.100 \text{ M} = 0.300 \text{ osmol/L}$ (isotonic). $\therefore$ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.

796. Chromic acid

797. 1.35×10^{-1} moles

798. $\text{S}_8 + 12 \text{O}_2 + 8 \text{H}_2\text{O} \rightarrow 8 \text{H}_2\text{SO}_4$

799. $5.899 \times 10^{27} \text{ g}$

800. No $\therefore$ only weak dispersion forces exist between O_2 and water, so solubility is low.

801. 0.120 M

802. $m = \frac{q}{\Delta H} = \frac{24.0 \times 10^3 \text{ J}}{334 \text{ J/g}} = 71.9 \text{ g}$

803. $\Delta H_{\text{f}} = \Delta H_{\text{sub}} + IE_1 + \frac{1}{2}D + EA + U = 89 + 419 + 121 + (-349) + (-717) = -437 \text{ kJ/mol}$

804. CO_2 is a molecular compound.

805. No — spontaneity is not the same as exothermicity. $\therefore$ dissolving scatters the ordered crystal lattice into freely moving hydrated ions, the large entropy increase outweighs the endothermic ΔH , so the process proceeds even while it cools the surroundings.

806. $\lambda = \frac{2d \sin \theta}{n} = \frac{2 \times 995 \times \sin(8.9^\circ)}{2} = 154 \text{ pm}$. $\therefore$ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d \sin \theta$, leading to $n\lambda = 2d \sin \theta$.

807. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. $\therefore$ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.

808. $\text{Cu}(\text{MnO}_4)_2 + 2 \text{NH}_4\text{I} \rightarrow \text{CuI}_2 + 2 \text{NH}_4\text{MnO}_4$

809. 49 mL

810. 313 K

811. Glycerol $\therefore$ glycerol's three O-H groups give far more hydrogen bonding than ethanol's one; stronger intermolecular forces increase viscosity.

812. $1s^2 2s^2$; Be forms Be^{2+} and is isoelectronic with He $\therefore$ given the neutral Be atom's electron configuration, it loses 2 electrons to match the noble-gas electron configuration of He.

813. $3.42 \times 10^5 \text{ Pa}$ $\therefore$ Dalton's law gives $P_{\text{gas}} = x_{\text{gas}}P_{\text{total}}$, so dividing the partial pressure by the mole fraction gives the total pressure before converting to SI units.

814. Boiling needs far more $\therefore$ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all intermolecular attractions to separate molecules completely.

815. Cl_2O_7

816. Ideal behavior $\therefore$ both are similar nonpolar hydrocarbons with comparable London dispersion forces $\rightarrow$ ideal behavior.

817. $1s^2 2s^2 2p^6 3s^2 3p^3$; P forms P^{3-} and is isoelectronic with Ar $\therefore$ given the neutral P atom's electron configuration, it gains 3 electrons to match the noble-gas electron configuration of Ar.

818. 3.16 g

819. 5.6 mg

820. $\theta = \arcsin\left(\frac{n\lambda}{2d}\right) = \arcsin\left(\frac{1 \times 154}{2 \times 200}\right) = 22.6^\circ$. $\therefore$ the angle determines the path difference between rays reflected from adjacent planes. The maximum occurs only when that path difference equals $n\lambda$, giving $n\lambda = 2d \sin \theta$.

821. $\text{Mg}^{2+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightarrow \text{MgS}(\text{s})$

822. $1.01 \times 10^1 \text{ mol}$

823. The $2s$ – $2p$ energy gap is larger in O, F, and Ne $\therefore$ increasing effective nuclear charge stabilizes the more penetrating $2s$ orbital more than the $2p$ orbital. The energy gap is therefore too large for the σ_{2s} and σ_{2p} orbitals to mix; in Li–N the gap is small

enough that mixing occurs, pushing the π_{2p} orbitals below σ_{2p} .

824. 1.083 g

825. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$

826. 0.0614 J/g/K

827. $3.5 \times 10^5 \text{ Pa}$. $\therefore$ the amount of gas is constant, so PV/T remains constant. Increasing the absolute temperature increases pressure, whereas increasing volume decreases pressure; both effects combine according to $P_1V_1/T_1 = P_2V_2/T_2$, and temperatures must be expressed in kelvin because the proportionality is with absolute temperature.

828. -342.1 kJ . Since $\Delta H < 0$, the reaction is exothermic.

829. Chromium (III) bisulfite dihydrate: $\text{Cr}(\text{HSO}_3)_3 \cdot 2\text{H}_2\text{O}$

830. $2.920 \times 10^1 \text{ MPa}$

831. 940. kg

832. 52.4% $\therefore$ 1 sphere of radius $r = a/2$; sphere vol = $(4/3)\pi r^3$; cell vol = a^3 ; efficiency = $(4/3)\pi(a/2)^3/a^3 = \pi/6 \approx 52.4\%$.

833. $2.1 \times 10^4 \text{ Pa}$. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.

834. $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$. $\therefore$ bond length is inversely related to bond order. Calculating the MO bond orders gives O_2^+ : 2.5, O_2 : 2.0, O_2^- : 1.5, O_2^{2-} : 1.0, so the species with the highest bond order has the shortest bond.

835. $8.42 \times 10^1 \text{ g}$

836. 10 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.

837. $a = \frac{4r}{\sqrt{3}} = \frac{4 \times 137}{\sqrt{3}} = 316.4 \text{ pm}$

838. Random error $\therefore$ the measurements fluctuate unpredictably in both directions.

839. Approximately zero $\therefore$ at very large separation the atoms are essentially noninteracting, and this state is defined as the zero-potential-energy reference.

840. 22 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:2 mole ratio, so the gas volumes must change in the same 1:2 ratio.

841. Yes, Tyndall effect is observed. Colloid $\therefore$ proteins and cells produce a Tyndall effect; plasma proteins are colloidal.

842. London dispersion $\therefore$ nonpolar homonuclear diatomic.

843. HClO_4

844. Oxidized: Mg Reduced: Ni

845. 3.05 m

846. Yes, iron and bromine form an ionic compound with the empirical formula FeBr_3 .

847. Increases. $\therefore$ increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.

848. 4 $\therefore$ 8 corner atoms $\times$ 1/8 + 6 face atoms $\times$ 1/2.

849. $1.60 \times 10^4 \text{ Pa}$. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.

850. Chemical property. The ability to rust of iron is a chemical property; it describes iron's tendency to react with oxygen and water to form iron oxide, a new substance.

851. Solubility increases by a factor of 3 $\therefore C = K_H P$; $C_1 = 7.8 \times 10^{-4} \text{ mol/L}$, $C_2 = 2.3 \times 10^{-3} \text{ mol/L}$; $C_2/C_1 = 3.0/1.0 = 3$.

852. Molecular solid (I_2) $\therefore$ molecular crystal: neutral I_2 molecules held by LDF; no free charges.

853. K_2O $\therefore$ larger ion(s) $\rightarrow$ weaker electrostatic attraction; see Coulomb's law ($F \propto \frac{q_1 q_2}{r^2}$)

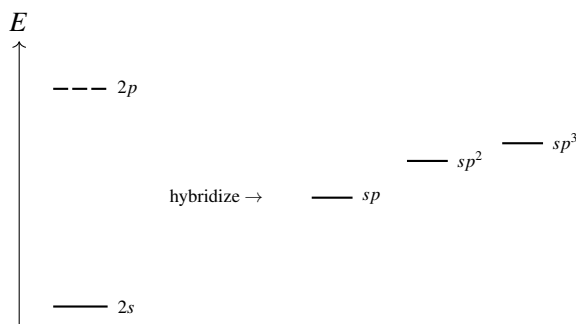
854. 62 kPa. $\therefore$ the gas sample and its temperature do not change, PV is constant. The smaller final volume produces a proportionally larger pressure, so the initial pressure must be lower by the inverse volume ratio.

855. $\text{KCl} < \text{NaCl} < \text{CaO}$ $\therefore$ L.E. increases with greater ionic charge and smaller ionic radius (Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$)

856. Solubility increases by a factor of 4 $\therefore C = K_H P$; $C_1 = 1.6 \times 10^{-3} \text{ mol/L}$, $C_2 = 6.2 \times 10^{-3} \text{ mol/L}$; $C_2/C_1 = 8.0/2.0 = 4$.

857. $2\text{HCl} + \text{Pb}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{PbCl}_2$

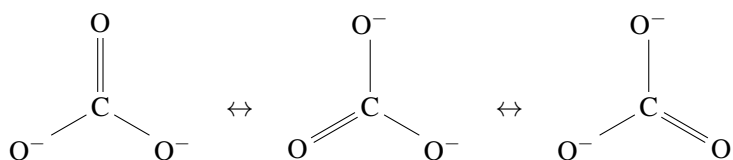
858. Tetrahedral holes, coordination number 4 $\therefore r_+/r_- = 45/140 = 0.32$ lies in the 0.225–0.414 range, which geometrically favors tetrahedral coordination.
859. Element
860. 4.146 g
861. $n_{\text{solute}} = \frac{25.0}{60.06} = 0.416 \text{ mol}$. $m = \frac{0.416 \text{ mol}}{0.100 \text{ kg}} = 4.16 \text{ mol/kg}$
862. $5.5 \times 10^2 \text{ g}$
863. (shortest wavelength) violet light < blue light < yellow light < infrared light (longest wavelength)
864. S
865. 4.18 g
866. $m = \frac{\Delta T_b}{i K_b} = \frac{0.768}{1 \times 0.512} = 1.50 \text{ mol/kg}$; $n = m \times m_{\text{solvent}} = 1.50 \times 1.00 \text{ kg} = 1.50 \text{ mol}$; $m_{\text{solute}} = 1.50 \text{ mol} \times 180.16 \text{ g/mol} = 270. \text{ g}$
867. $2.62 \times 10^{-6} \text{ m}$ $\therefore \Delta E = \left(\frac{-R_y}{n_2^2} \right) - \left(\frac{-R_y}{n_1^2} \right)$ and $\Delta E = \frac{hc}{\lambda}$
868. 5.2 mm
869. $\text{HC}_2\text{H}_3\text{O}_2$
870. Compound
871. Pure substance. It is a pure substance (compound); it has a definite molecular formula and a fixed composition.
872. 1.30 mol
873. sp^3 $\therefore$ it contains the greater fraction of p character. Because the $2p$ orbitals are higher in energy than the $2s$ orbital, increasing the contribution of $2p$ orbitals raises the weighted-average expectation value, whereas increasing s character lowers it.



874. 0.96 atm; No. A liquid boils only when its vapor pressure equals or exceeds the external pressure, allowing vapor bubbles to remain stable throughout the liquid. Since the vapor pressure is less than the atmospheric pressure, vapor bubbles cannot continue to grow.
875. $4\text{O}_2 + 3\text{H}_2 + \text{N}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{HNO}_3$
876. 10 $\therefore$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
877. $2.4 \times 10^{-3} \text{ mol}$
878. $\text{HCl}(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{Cl}^-(aq) + \text{H}_3\text{O}^+(aq)$
879. 19 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:2 mole ratio, so the gas volumes must change in the same 1:2 ratio.
880. 0.284 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
881. The liquid and vapor densities converge and become equal $\therefore$ as T rises the liquid expands and the vapor densifies until, at the critical point, the two are identical and indistinguishable.
882. 8.67 g
883. No $\therefore$ a σ bond forms from head-on overlap and provides the primary connection between the nuclei. It is energetically most favorable to form, so it will form before a π bond forms. A π bond forms from side-by-side overlap of parallel orbitals and can exist only in addition to an existing σ bond.
884. 712 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined

- by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
885. Colloidal particles carry like surface charges that repel one another and stay dispersed; the added ions neutralize those charges, removing the repulsion so the particles aggregate and settle out.
886. $4 \times \frac{1}{8} + 6 \times \frac{1}{2}$
887. $-1.69 \times 10^5 \text{ J}$
888. Increases. $\therefore$ at the same temperature and pressure every ideal gas has the same molar concentration, but each mole of the replacement gas has a larger mass. More mass therefore occupies the same volume.
889. The orientation with greater orbital overlap produces the stronger bond $\therefore$ increased overlap concentrates more electron density between the nuclei, increasing the attractive interaction between the nuclei and the shared electrons and lowering the energy of the bonded state.
890. Physical change $\therefore$ the pencil changes size and shape but the materials remain the same.
891. N-butane $\therefore$ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.
892. $5.64 \frac{\text{g}}{\text{mL}}$
893. Copper (II) oxalate tetrahydrate: $\text{CuC}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$
894. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
895. Cation:anion = 1 : 1 $\therefore$ counting the ions per unit cell from a face-centered cubic array of anions with cations in half of the tetrahedral holes gives that ratio (the zinc blende (ZnS) structure).
896. Oxidized: Rb Reduced: W
897. ionic ($\Delta\text{EN} = 3.0$); F carries the δ^- $\therefore \Delta\text{EN} \geq 2.0$ and F is the more electronegative atom
898. $\rho = \frac{ZM}{N_A a^3} = \frac{2 \times 55.845}{6.022 \times 10^{23} \times (286.65 \times 10^{-10})^3} = 7.87 \text{ g/cm}^3$. $\therefore$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.
899. 60
900. The SI base unit for amount of substance is the mole (symbol: mol).
901. $1.0 \times 10^2 \text{ g/mol}$. $\therefore$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
902. Ideal behavior $\therefore$ both are largely nonpolar molecules with similar intermolecular forces $\rightarrow$ nearly ideal solution.
903. 1.3 g. $\therefore$ the pressure, volume, and temperature determine the number of moles present from the ideal gas law, multiplying that amount by the molar mass gives the sample mass.
904. It will sublime $\therefore$ the external pressure (0.0225 atm) is below the triple-point pressure (0.12 atm), so no liquid is stable and the solid converts directly to gas.
905. 0.05778 M
906. 8 molecular orbitals $\therefore$ eight atomic orbitals participate, so eight molecular orbitals are formed.
907. $n_{\text{ace}} = \frac{9.21 \text{ g}}{58.08} = 0.159 \text{ mol}$; $n_{\text{H}_2\text{O}} = \frac{50.0 \text{ g}}{18.015} = 2.78 \text{ mol}$; $\chi_{\text{ace}} = 0.0540$; $\chi_{\text{H}_2\text{O}} = 0.946$; $\chi_{\text{ace}} + \chi_{\text{H}_2\text{O}} = 1.00 \checkmark$
908. Emulsion $\therefore$ it is a liquid dispersed in a liquid.
909. sp^2 for the carbonyl C; sp^3 for the two methyl carbons. The carbonyl C has 3 electron groups (2 C–C + 1 C=O), giving sp^2 . Each methyl C has 4 single bonds, giving sp^3 $\therefore$ electron groups determine hybridization.
910. 8.6 mg
911. $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
912. Basis: 100 g solution $\rightarrow$ 25.0 g glucose, 75.0 g water; $n_{\text{solute}} = \frac{25.0 \text{ g}}{180.16 \text{ g/mol}} = 0.139 \text{ mol}$; $m = \frac{0.139 \text{ mol}}{0.0750 \text{ kg}} = 1.85 \text{ mol/kg}$
913. 0.2913 moles
914. $198.44 \frac{\text{g}}{\text{mol}}$
915. 3.9×10^{22} oxygen atoms
916. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$

917. 26 mL. $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{O}_2:\text{Hg}$ is 1:2 so the volume changes by the same factor.
918. The molecular orbital with less electron density between the nuclei has the higher energy $\therefore$ electron density located between the nuclei is attracted to both nuclei simultaneously, increasing the stabilizing electrostatic attraction. Greater density between the nuclei therefore lowers the total energy, whereas less density leaves the nuclei less effectively bound and the energy higher.
919. Supersaturated $\therefore$ unstable; crystallization occurs on seeding or disturbance.
920. $\text{E} \rightarrow \text{D}$ $\therefore$ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
921. $7.8 \frac{\text{g}}{\text{mL}}$
922. 0.62. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
923. No $\therefore$ benzene cannot form favorable polar or hydrogen-bonding interactions with water.
924. 270 zJ $\therefore$ "C" is the level with the third-lowest energy, AKA the second excited state.
925. Obeys the octet rule $\therefore$ the central N atom is surrounded by 8 electrons
926. Solution A: $\Delta T = i K m = 2 \times 1.86 \times 0.200 = 0.744^\circ\text{C}$; Solution B: $\Delta T = 1 \times 1.86 \times 0.200 = 0.372^\circ\text{C}$. Solution A has the greater freezing point depression $\therefore$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).
927. 104°F
928. $n_{\text{solute}} = \frac{30.0}{342.3} = 0.0876 \text{ mol}$; $n_{\text{water}} = \frac{100.0}{18.015} = 5.55 \text{ mol}$. $\chi_{\text{solute}} = \frac{0.0876}{0.0876 + 5.55} = 0.0155$
929. $3.15 \times 10^1 \text{ cm}$
930. $\text{H}_2\text{CrO}_4 + \text{Mg}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{MgCrO}_4$
931. $\Delta H_{\text{vap}} = 2260 \text{ J/g} \times 18.015 \text{ g/mol} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 40.7 \text{ kJ/mol}$
932. Per 1 L of solution: mass solution = $1.02 \times 10^3 \text{ g}$; mass solute = $0.500 \text{ mol} \times 58.44 \text{ g/mol} = 29.2 \text{ g}$; mass solvent = $1.02 \times 10^3 \text{ g} - 29.2 \text{ g} = 996 \text{ g}$; $m = \frac{0.500 \text{ mol}}{0.996 \text{ kg}} = 0.502 \text{ mol/kg}$
933. Element $\therefore$ it contains only aluminum atoms.
934. C_2 $\therefore$ bond order $2 > 1$ for B_2 ; a higher bond order means a stronger and more stable bond.
935. -10 J
936. $\text{CH}_3\text{OH} > \text{C}_2\text{H}_5\text{OH} > \text{C}_5\text{H}_{11}\text{OH} > \text{C}_6\text{H}_{13}\text{OH}$ $\therefore$ all share the same H-bonding $-\text{OH}$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.
937. 80%
938. 0.000070 m
939. $\text{B} \rightarrow \text{A}$ $\therefore$ because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).
940. ionic and covalent $\therefore$ the compound contains polyatomic ion(s). The atoms within each polyatomic ion are held together by covalent bonds, while the polyatomic ions and any oppositely charged ions are held together by ionic bonds.
- 941.



the bond order of each C–O bond is $4/3$ or 1.33 $\therefore$ the bond order of a bond in a resonance hybrid is the average of its bond orders in the equivalent resonance structures; all of the resonance structures contribute equally, so every C–O bond is identical in the resonance hybrid. Sum the bond order of each C–O bond in CO_3^{2-} and divide by the number of equivalent resonance structures.

942. N_2 $\therefore$ bond order $3 > 2$ for O_2 ; higher bond order means stronger, shorter, and more stable bonding.
943. Silver sulfide: Ag_2S
944. The substance with the deeper well $\therefore$ a deeper well means stronger attractions, so more thermal energy is required to disrupt its lattice and melt it.

945. 55.6%

946. $M = \frac{\Pi}{RT} = \frac{7.60 \times 10^{-3} \text{ atm}}{0.08206 \times 298 \text{ K}} = 3.11 \times 10^{-4} \text{ mol/L}$; $\mathcal{M} = \frac{m/V}{M} = \frac{20.0 \text{ g/L}}{3.11 \times 10^{-4} \text{ mol/L}} \approx 6.44 \times 10^4 \text{ g/mol}$

947. No $\therefore$ the π electrons are delocalized equally over all six carbon atoms rather than confined to three individual double bonds. Equal electron sharing gives every carbon-carbon bond the same average bond order, so all six bonds have the same length.

948. 13 protons and 14 neutrons.

949. $\text{O}=\text{O} < \text{O}-\text{O}$ $\therefore$ higher bond order means more shared electron pairs between the nuclei, increasing the attraction between the atoms. Stronger attraction pulls the atoms closer together, so bond length decreases as bond order increases.

950. Hybridization: sp^3 . Electron-Group Geometry: tetrahedral. Molecular Geometry: tetrahedral.
 $\therefore$ there are 4 bonds and 0 lone pairs around the central atom, there are 4 total electron groups surrounding it and $\therefore$ each requires one hybrid orbital, the central atom is sp^3 hybridized. The electron-group geometry is tetrahedral. Because all electron groups are bonding pairs, the molecular geometry is also tetrahedral with bond angles of 109.5° .

951. AgF and H_2O

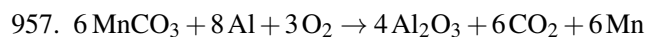
952. ionic; high melting point $\therefore$ its ions are held by strong electrostatic attractions throughout the lattice

953. ionic $\therefore$ it consists of a metal and a nonmetal, so the particles are held together by ionic bonds.

954. 1.94×10^{-56} moles

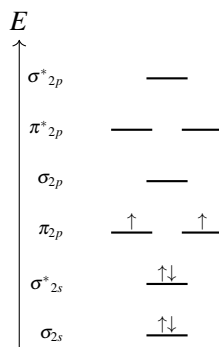
955. 88.19 mL

956. $\Delta H_{\text{vap}} = 83.5 \text{ kJ/mol}$ $\therefore -R \frac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)



958. AgBr is an ionic compound.

959. Bond order = $\frac{4-2}{2} = 1$; paramagnetic (2 unpaired electrons). There are 6 valence electrons. The molecular orbitals are ordered using the correct mixing model and populated according to the Aufbau principle, the Pauli exclusion principle, and Hund's rule. The bond order follows from $\frac{N_b - N_a}{2}$, and the magnetic behavior follows from the presence of unpaired electrons in the completed molecular-orbital diagram.



960. sp^3 $\therefore$ each π bond requires one unhybridized p orbital, so 0 possible π bonds correspond to 0 unhybridized p orbitals, meaning 3 of the three p orbitals were hybridized together with one s .

961. $a = 2\sqrt{2}r = 2\sqrt{2} \times 128 = 362.0 \text{ pm}$

962. $7.50 \times 10^1 \text{ fm}$

963. $1.25 \times 10^3 \text{ mol}$. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .

964. No. $\therefore$ an isolated atom is described by its atomic orbitals; hybrid orbitals are a model constructed to describe the bonding environment, so they are not treated as pre-existing physical objects before bonding.

965. Molecular solid $\therefore$ its composition implies an extremely low melting point, softness, and no conduction.

966. $207.9 \frac{\text{g}}{\text{mol}}$

967. 0.179 g

968. $\text{O}=\text{O} < \text{O}-\text{O}$ $\therefore$ higher bond order means more shared electron pairs between the nuclei, increasing the attraction between the atoms. Stronger attraction pulls the atoms closer together, so bond length decreases as bond order increases.

969. Boiling needs far more $\therefore$ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all

intermolecular attractions to separate molecules completely.

970. LiCl $\therefore$ greatest boiling point elevation $\therefore$ largest effective molality. $\text{C}_6\text{H}_{12}\text{O}_6$: $i \times m = 1 \times \frac{1.00}{180.16} = 5.55 \times 10^{-3} \text{ mol/kg}$; AlCl_3 :
 $i \times m = 4 \times \frac{1.00}{133.34} = 0.0300 \text{ mol/kg}$; LiCl : $i \times m = 2 \times \frac{1.00}{42.39} = 0.0472 \text{ mol/kg}$.
971. the C–O bond in ethanol ($\text{C}_2\text{H}_5\text{OH}$). $\therefore$ the Lewis structure, including any resonance contribution where applicable, must first be used to determine the *effective amount of bonding electron density shared between the nuclei*. Smaller bonding electron density creates a shallower potential-energy well, so less energy is required to separate the atoms completely.
972. sp^2 for the carbonyl C; sp^3 for the two methyl carbons. The carbonyl C has 3 electron groups (2 C–C + 1 C=O), giving sp^2 . Each methyl C has 4 single bonds, giving sp^3 $\therefore$ electron groups determine hybridization.
973. Increases. $\therefore$ increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.
974. Oxidizing Agent: H_2 Reducing Agent: Te
975. $C = K_H P = 1.4 \times 10^{-3} \times 4.0 \text{ atm} = 5.6 \times 10^{-3} \text{ mol/L}$
976. 1s, 2s, 2p, 4p, 5s
977. 87.6 μg
978. -923 kJ $\therefore$ Hess's law: $\Delta H_f^\circ = \Delta H_{\text{sub}} + IE + \Delta H_{\text{diss}} + EA + \Delta H_{\text{lat}}^\circ$
979. $q = n\Delta H_{\text{vap}} = \frac{14.4 \text{ g}}{32.042 \text{ g/mol}} \times 35.2 \text{ kJ/mol} = 15.8 \text{ kJ}$
980. Per 1 L of solution: mass solution = $1.02 \times 10^3 \text{ g}$; mass solute = $0.500 \text{ mol} \times 58.44 \text{ g/mol} = 29.2 \text{ g}$; mass solvent = $1.02 \times 10^3 \text{ g} - 29.2 \text{ g} = 996 \text{ g}$; $m = \frac{0.500 \text{ mol}}{0.996 \text{ kg}} = 0.502 \text{ mol/kg}$
981. Amorphous $\therefore$ its random, disordered structure looks the same in all directions and softens over a range, unlike an ordered crystal, which can be anisotropic and melts sharply.
982. 55°C
983. $6.62 \times 10^{-7} \text{ m}$ $\therefore$ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$
984. No new substance is formed $\therefore$ The wire changes shape but remains copper.
985. 2.9 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
986. N_2 $\therefore$ bond order $3 > 2$ for O_2 ; higher bond order means stronger, shorter, and more stable bonding.
987. Answer: X belongs to Group 17 $\therefore$ X is F-like and has 7 valence electrons.
988. No layers form $\therefore$ ethanol is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.
989. $F = 1 - 3 + 2 = 0$. $\therefore$ Neither temperature nor pressure can be changed independently while all three phases remain in equilibrium because the simultaneous phase equilibria completely constrain the system.
990. $\text{Fe}_2\text{O}_3 + 3 \text{CO} \rightarrow 2 \text{Fe} + 3 \text{CO}_2$
991. Water flows out of the cell; the cell shrivels (crenation). $\therefore$ the solution is hypertonic: particle concentration = $1 \times 0.500 \text{ M} = 0.500 \text{ osmol/L} > 0.300 \text{ osmol/L}$ inside.
992. $T_2 = \left(\frac{1}{T_1} - \frac{R \ln(P_2/P_1)}{\Delta H_{\text{vap}}} \right)^{-1} = 340.1 \text{ K} = 66.9^\circ\text{C}$ $\therefore$ boiling occurs when the vapor pressure equals the external pressure, and a lower external pressure lowers the boiling point.
993. $2.26 \times 10^4 \text{ Pa}$. $\therefore$ the balanced equation determines the amount of H_2 produced from the amount of HCl that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
994. $3.716 \frac{\text{g}}{\text{mL}}$
995. $\text{Cr}_2(\text{SO}_3)_3$ and H_2O
996. No $\therefore$ every carbon must contribute a parallel unhybridized p orbital to create continuous overlap around the ring. Removing one p orbital breaks the continuous overlap, preventing delocalization over all six carbon atoms.
997. -13°F

998. $\rho = \frac{ZM}{N_A a^3} = \frac{4 \times 58.693}{6.022 \times 10^{23} \times (352.40 \times 10^{-10})^3} = 8.91 \text{ g/cm}^3$. $\therefore$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.
999. $5\text{O}_2 + \text{P}_4 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
1000. $\text{Li}_2 = \text{B}_2$ (BO=1) < N_2 (BO=3). Individual bond orders: Li_2 : BO=1, B_2 : BO=1, N_2 : BO=3. $\therefore$ bond order depends on the balance between electrons in bonding and antibonding molecular orbitals: electrons added to bonding orbitals increase bond order, whereas electrons added to antibonding orbitals decrease it. A complete justification correctly accounts for the MO occupancy of each species and uses it to explain the ranking, not merely state the bond-order values.